

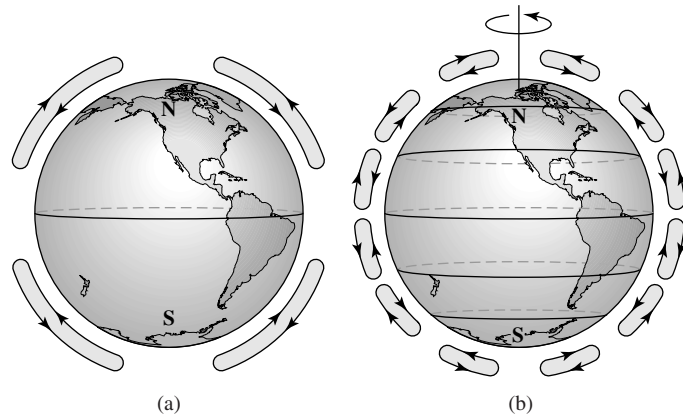


FIGURE 19.7 (a) The pattern of hypothetical Hadley circulation, caused by warm air rising near the equator and cool air sinking near the poles. (b) General weather circulation on Earth caused by the Hadley cells being broken up by radiative cooling of the warm air migrating toward the poles.

latitude L is just

$$v_L = \omega r = \omega R_{\oplus} \cos L.$$

At the equator ($L = 0^\circ$), this speed is approximately $465 \text{ m s}^{-1} = 1670 \text{ km h}^{-1}$. However, at a latitude of $L = 40^\circ$, the speed is reduced to 1300 km h^{-1} . As a result, according to an observer in an inertial reference frame, a person standing still on the surface of Earth at the equator is moving approximately 370 km h^{-1} faster than someone standing still on the surface at a latitude of 40° . This velocity difference with latitude affects weather circulation patterns.

As an illustration of the effect of the Coriolis force, consider the apparent motion of a projectile fired nearly horizontally from the equator northward, as shown in Fig. 19.8(a). Also assume that the elevation of the projectile above the surface of Earth is essentially constant during its flight. From the point of view of an observer on Earth's surface located at the origin of the projectile's motion, the projectile *initially appears* to be traveling straight north since the observer has the same easterly speed as the easterly component of the projectile's velocity vector at the time of launch. However, from the point of view of an observer in an inertial reference frame far above the planet, the direction of the projectile's motion will be northeast precisely because it has an eastward component to its velocity vector.

As the projectile travels north, it will appear to an observer on the ground that the path will deflect toward the east due to some undetected “force” [Fig. 19.8(b)]. However, the observer in an inertial frame understands this observation as being due to the velocity difference of Earth's surface with latitude; the easterly component of the projectile's velocity vector will cause it to “outrun” observers at progressively more northerly latitudes on Earth.

It can be shown that the value of the Coriolis force as measured in a noninertial reference frame fixed to the surface of Earth is given by

$$\boxed{\mathbf{F}_C = -2m\boldsymbol{\omega} \times \mathbf{v},} \quad (19.11)$$

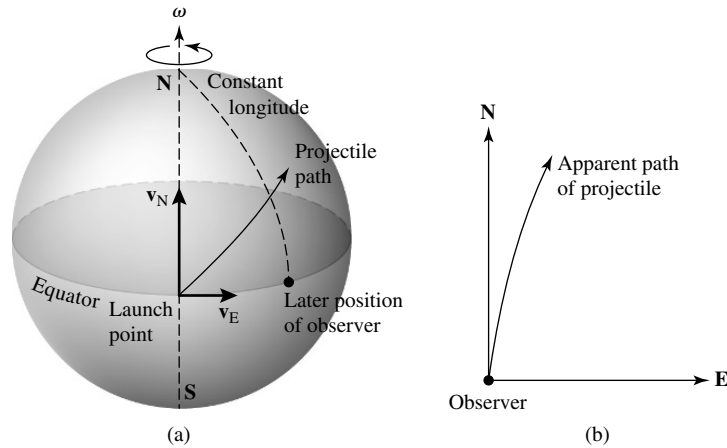


FIGURE 19.8 A projectile launched from the equator northward will curve toward the east as seen by an observer on Earth. (a) The view from an observer in an inertial frame above Earth. (b) The apparent trajectory as seen from an observer at rest on the surface of Earth.

where ω is the angular velocity vector of the planet and $\mathbf{v}$ is the velocity of the projectile with respect to the noninertial reference frame. Clearly, as the velocity of the particle or the angular velocity of the planet increases, the effect of the “force” increases. On Earth, the presence of the Coriolis force causes the large-scale north–south circulation patterns to develop global east–west zonal flows [recall Fig. 19.7(b)]. Nearest the equator, the circulation patterns are generally easterly and are known as the trade winds. In the zonal regions between about 30° and 55° , the prevailing winds are the westerlies, and near the poles, the flow is again generally easterly. The Coriolis force is also responsible for the motions of clouds around high- and low-pressure systems (see Problem 19.21).

The Complexities of Weather Systems

As anyone who has watched weather forecasts knows, weather circulation patterns on Earth are much more complex than has been described above. Effects such as moisture in the atmosphere, the diversity of land forms, the transport of heat by ocean currents, temperature differences between oceans and land, and even frictional effects between the atmosphere and the surface of the planet all contribute to the complexity of Earth’s weather systems.

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PROBLEMS

- 19.1** (a) Based on the data given in Appendix C, express the masses of the Moon, Io, Europa, Ganymede, Callisto, Titan, Triton, and Pluto in units of the mass of Mercury.
- (b) Express the radii of these moons and Pluto in units of the radius of Mercury.
- 19.2** A second version of Bode's rule (the Blagg–Richardson formulation) is given by

$$r_n = r_0 A^n,$$

where n is the number of the planet in order from the Sun outward (e.g., $n = 1$ for Mercury) and r_0 and A are constants.

- (a) Plot the position of each planet and that of Ceres on a semilog graph of $\log_{10} r_n$ vs. n .
- (b) Draw the best-fit straight line through the data on your graph and determine the constants r_0 and A .
- (c) Compare the “predictions” of your fit with the actual values for each planet by calculating the relative error,

$$\frac{r_n - r_{\text{actual}}}{r_{\text{actual}}}.$$

- 19.3** Repeat Problem 19.2 for the Galilean moons of Jupiter (Io, Europa, Ganymede, and Callisto). Express their orbital distances in units of the radius of Jupiter. The data for Jupiter and its moons are found in Appendix C.
- 19.4** Starting from Eq. (19.2) and using the geometry in Fig. 19.3, derive Eq. (19.3).
- 19.5** (a) Assuming for simplicity that Earth is a sphere of constant density, compute the rate of change in rotational angular momentum of Earth due to the tidal influence of the Moon. Is the change positive or negative?
 (b) Treating the Moon as a point mass, estimate the rate of change in orbital angular momentum of the Moon. Is this change positive or negative?
 (c) Comparing your crude answers to parts (a) and (b), what can you say about the total angular momentum of the Earth–Moon system over time?
- 19.6** (a) Make a rough estimate of how long it will take for Earth’s rotation period to reach 47 days, at which time it will be synchronized with the Moon’s orbital period.
 (b) Based on what you know about the evolution of our Sun, will future inhabitants of Earth ever get the opportunity to see the Earth–Moon system completely synchronized (Earth always keeping the same “face” toward the Moon)? Why or why not?
- 19.7** (a) Using Kepler’s laws, estimate the distance of the Moon from Earth at some time in the distant future when the Earth–Moon system is completely synchronized at 47 days.
 (b) As seen from Earth, what will the angular diameter of the Moon be at that time?
 (c) Assuming that the Sun’s diameter is the same as the present-day value, would a total eclipse of the Sun be possible? Why or why not?
- 19.8** (a) Calculate the ratio of the tidal forces on Earth due to the Moon and the Sun.
 (b) With the aid of vector diagrams, explain the cause of the strong spring tides and the relatively weak neap tides.
- 19.9** Explain the almost complete lack of any tides in the Arctic Ocean at the latitude of Barrow, Alaska (71.3° N).
- 19.10** Using the data in Appendix C, estimate the Roche limit for the Mars–Phobos system. Phobos’s mean density is 2000 kg m^{-3} and it orbits at a distance of $9.4 \times 10^6 \text{ m}$. Explain the suggestion that Mars may develop a small ring system in the future.
- 19.11** Why aren’t spacecraft tidally disrupted when they pass near the giant planets?
- 19.12** Including rotation, rederive Eq. (19.4) for the case of a spherical moon in synchronous rotation about a planet. What is your new value for f_R ? *Hint:* You may find Kepler’s third law helpful.
- 19.13** (a) Use Eq. (3.17) and simple geometry to derive Eq. (19.5) for the temperature T_p of a planet at a distance D from the Sun.

- (b) Imagine the greenhouse gases in Earth's atmosphere to be a single layer that is completely transparent to the visible wavelengths of light received from the Sun, but completely opaque to the infrared radiation emitted by the surface of Earth. Assume that the top and bottom surface areas of the layer are each equal to the surface area of the planet and that the temperature at the top of this atmospheric layer, $T_{\oplus}$, is just the blackbody temperature found in Example 19.3.1. Show that the blackbody radiation emitted by this atmospheric layer results in a warming of Earth's surface to a temperature of $T_{\text{surf}} = 2^{1/4} T_{\oplus}$. Compare this result with Earth's average surface temperature of $15^{\circ}\text{C} = 59^{\circ}\text{F}$.
- 19.14** Using Eq. (19.6), estimate the temperature that would be required for all of the *atomic* hydrogen to escape Earth's atmosphere. Is this consistent with the lack of significant amounts of atomic or molecular hydrogen in the atmosphere? Why or why not?
- 19.15** (a) Estimate the equilibrium blackbody temperature of Jupiter. Use the data found in Appendix C.
 (b) Using Eq. (19.6), estimate the temperature that would be required for all of the hydrogen molecules to escape Jupiter's atmosphere since the planet's formation.
 (c) Based on your answer in part (b), what would you expect the dominant component of the atmosphere to be? Why?
- 19.16** Using integration by parts, show that Eqs. (19.8) and (19.9) follow directly from Eq. (19.7).
- 19.17** Taking the density of air to be 1.3 kg m^{-3} near the surface of Earth, show that the number density of nitrogen molecules is approximately $2 \times 10^{25} \text{ m}^{-3}$, as given in Example 19.3.3.
- 19.18** Assuming that the mean free path of molecules in Earth's exosphere is sufficiently long ($\sim 500 \text{ km}$) to allow them to escape into interplanetary space, use Eq. (9.12) to estimate the number density of molecules in the exosphere. Note that you will need to make an order-of-magnitude estimate of their collision cross sections. Compare your result with the number density of nitrogen molecules quoted in Example 19.3.3. Explain any significant differences between the two values.
- 19.19** (a) Suppose that Earth once had an atmosphere composed entirely of hydrogen atoms, rather than the molecular nitrogen and oxygen of today. Using Eq. (19.9), calculate the atmospheric escape parameter ν in this case if the temperature of the exosphere was 1000 K .
 (b) Using a procedure identical to that of Problem 19.18, estimate the number density of hydrogen atoms in the primordial exosphere.
 (c) What would have been the rate of loss of hydrogen atoms from the exosphere?
 (d) Assume that the number of atomic hydrogen atoms in the atmosphere was essentially the same as the number of nitrogen molecules today. Approximately how long would it take for the hydrogen to escape from the planet's atmosphere? Express your answer in years and compare it to the age of Earth. (*Note:* In Section 23.2 we will learn that it appears unlikely that Earth ever had a substantial hydrogen atmosphere.)
- 19.20** Calculate the atmospheric escape parameter ν for atomic hydrogen in Jupiter's exosphere (use $T \sim 1200 \text{ K}$). Compare your result with the value obtained for Earth (see Fig. 19.6 or the result of Problem 19.19a). *Hint:* Because of numerical limitations on most calculators, you may find it necessary to first determine $\log_{10} \nu$ rather than determining ν directly.
- 19.21** (a) Consider the case of a projectile launched from the North Pole toward the equator. With the aid of a diagram, show that the projectile is deflected westward (to the right as viewed from the launch point).

- (b) Recalling that a projectile launched from the equator toward the North Pole is also deflected toward the right (eastward), show that the circulation around low-pressure systems is counterclockwise in the Northern Hemisphere.
- (c) Which way do low-pressure systems circulate in the Southern Hemisphere?

19.22 Suppose that a ball of mass m is thrown with a velocity

$$\mathbf{v} = v_x \hat{\mathbf{i}} + v_y \hat{\mathbf{j}} + v_z \hat{\mathbf{k}},$$

where $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, and $\hat{\mathbf{k}}$ are unit vectors pointing directly east, north, and upward, respectively, at the point where the ball is thrown. The latitude of the ball is L when it is thrown.

- (a) Show that the components of the Coriolis force on the ball are given by

$$\mathbf{F}_C = -2m\omega[(v_z \cos L - v_y \sin L)\hat{\mathbf{i}} + v_x \sin L \hat{\mathbf{j}} - v_x \cos L \hat{\mathbf{k}}].$$

Hint: Be sure to represent the components of the vector $\boldsymbol{\omega}$ in terms of the coordinate system on the surface of Earth defined by $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, and $\hat{\mathbf{k}}$, with the origin of the system at the position where the ball was thrown.

- (b) What is the value of ω for Earth?
- (c) If the ball is thrown eastward with an initial velocity vector $\mathbf{v} = 30 \text{ m s}^{-1} \hat{\mathbf{i}}$ on the surface of Earth at a latitude of 40° , what are the components of the acceleration vector that are due to the Coriolis force?
- (d) If the ball is thrown northward with $\mathbf{v} = 30 \text{ m s}^{-1} \hat{\mathbf{j}}$, what are the components of the acceleration vector?
- (e) If the ball is thrown straight up with $\mathbf{v} = 30 \text{ m s}^{-1} \hat{\mathbf{k}}$, what are the components of the acceleration vector? Give a simple physical explanation for the result.

CHAPTER

20

The Terrestrial Planets

20.1	<i>Mercury</i>
20.2	<i>Venus</i>
20.3	<i>Earth</i>
20.4	<i>The Moon</i>
20.5	<i>Mars</i>

20.1 ■ MERCURY

The four terrestrial planets have a number of characteristics in common, such as being small, rocky, and slowly rotating (see Table 19.1). Our own Moon and several of the moons of the giant planets also share many of those same characteristics. In this chapter we shall focus our attention on the terrestrial planets and their moons, saving our discussion of the giant planets and their systems for Chapter 21.

The 3-to-2 Spin–Orbit Coupling of Mercury

As we learned in Section 17.1, the innermost planet, Mercury (Fig. 20.1), orbits so close to the Sun (0.39 AU) that Kepler’s laws begin to break down. The reason is that spacetime in the vicinity of massive objects is affected in such a way that Newton’s familiar inverse-square law (Eq. 2.11) is no longer a completely adequate description of gravity. It was the slow advance of the perihelion point of Mercury’s rather eccentric orbit ($e = 0.2056$) that presented one of the first tests of Einstein’s general theory of relativity.

The first hint that Mercury’s orbit also exhibits another curious feature came in 1965 when Rolf B. Dyce and Gordon H. Pettengill successfully bounced radar signals off the planet using the Arecibo radio telescope. The reflected signals had a spread of wavelengths that revealed Mercury’s rotation speed; because of the Doppler effect, radio waves that hit the approaching limb were blueshifted and those that struck the receding limb were redshifted. These observations indicated that Mercury’s rotation period was approximately 59 days. More precise measurements made by the **Mariner 10** spacecraft during its repeated flybys of the planet in 1974 and 1975 showed that the rotation period was actually 58.6462 days, exactly two-thirds the length of its sidereal orbital period of 87.95 days.

How this peculiar 3-to-2 relationship between rotation and orbital periods developed can be understood in light of the process of tidal evolution discussed in Section 19.2. At perihelion, Mercury experiences the strongest tidal force, causing the planet to try to align its bulge axis along the line connecting the planet’s center of mass to the center of mass of the

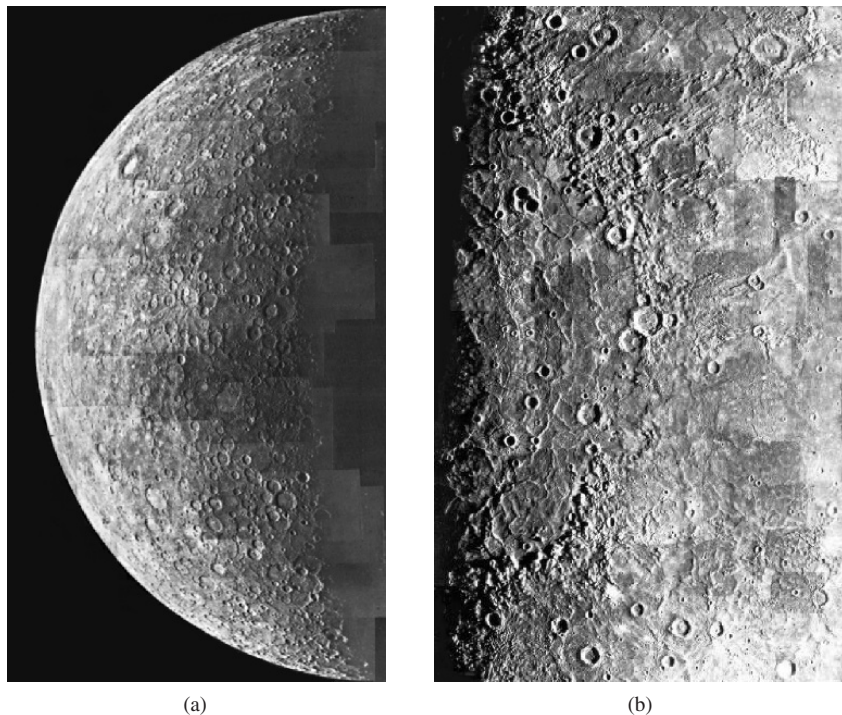


FIGURE 20.1 (a) Mercury, as seen by Mariner 10 when it was 200,000 km from the planet on March 29, 1974. (b) A portion of Caloris Basin can be seen near the terminator (the line separating day and night). Notice the semicircular rings of mountains centered on the impact point at the left-hand edge. (Courtesy NASA/JPL.)

Sun. As a result, due to the tremendous energy dissipation from friction that accompanies tidal distortions, Mercury's spin slowed to the point where the alignment ultimately did occur at perihelion during each orbit; see Fig. 20.2.

The Surface of the Planet

Pictures returned by Mariner 10 revealed a planet that bears a strong superficial resemblance to the Moon (compare Figs. 20.1 and 20.16). Mercury is a world that is heavily cratered, indicating that it underwent extensive bombardment during its nearly 4.6-billion-year history. Such evidence of violent collisions is commonplace on many worlds, giving us a hint of the Solar System's history. One impact (at what is now known as Caloris Basin) was so large that it created ripples that traveled across the planet and converged on the opposite side to produce a jumbled collection of hills.

A careful comparison of images of the Moon and Mercury shows that Mercury's craters are often separated by regions that are largely devoid of significant cratering. Assuming that the rate of impact was roughly the same on both worlds throughout their histories, and that they formed at approximately the same time, Mercury's surface must have been refreshed

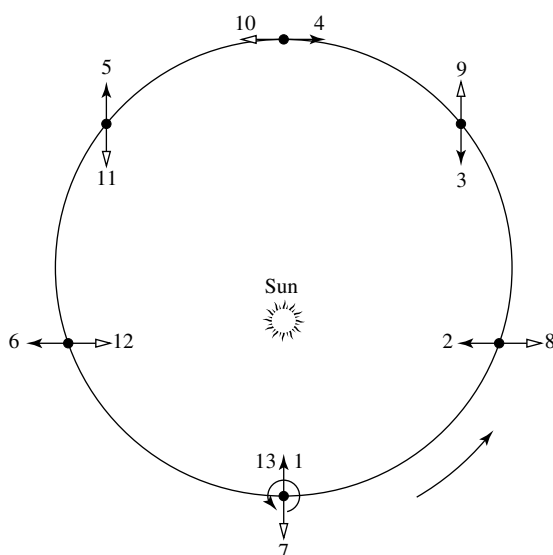


FIGURE 20.2 The 3-to-2 spin-orbit coupling of Mercury.

more recently (meaning it must be somewhat younger) than most of the Moon's surface. This is consistent with the conclusion that since Mercury is larger and closer to the Sun, it would have cooled off more slowly after formation, and hot, molten material would have been more likely to reach the surface to cover older impact sites.

Given the planet's size and its proximity to the Sun, it is not surprising that Mercury has only a very tenuous atmosphere (recall the analysis of the retention of planetary atmospheres in Section 19.3). Because of Mercury's high temperature on the subsolar side (reaching 825 K) and its relatively low escape velocity (4.3 km s^{-1}), atmospheric gases quickly evaporate into space. In fact, its exosphere reaches down to the surface of the planet. What atmosphere it does possess (number densities are less than 10^{11} m^{-3}) is due to charged nuclei of hydrogen and helium from the strong solar wind that become trapped in its weak magnetic field, together with atoms of oxygen, sodium, potassium, and calcium that have escaped the surface **regolith** (or soil) of the planet. The atoms that leave the regolith may have been liberated by impacting solar wind particles or the vaporization of regolith material by micrometeorites.

Ironically, radar data suggest that this closest planet to the Sun possesses highly reflective volatile material, probably water ice, in permanently shadowed craters near the polar caps.¹ Because tidal interactions have forced the planet's rotation axis to be almost exactly perpendicular to its orbital plane, the polar regions never get more than a very small amount of sunlight. Moreover, with virtually no atmosphere to speak of, Mercury cannot efficiently transport heat away from the equatorial regions. As a result, temperatures near the poles probably never exceed 167 K, and in shadows within craters near the poles, temperatures

¹The NASA 70-m tracking station at Goldstone, California, was used to send the nearly 500-kW signal at a wavelength of 3.5 cm, and the VLA received the reflected beam.

may be as low as 60 K. These temperatures are low enough that any water ice that may have been deposited there by a process such as cometary collisions would sublime only over very long time scales.

The Interior

Mercury's relatively high average density (5427 kg m^{-3}), when compared to that of the Moon (3350 kg m^{-3}), indicates that it must have lost most of its lighter elements and undergone enough gravitational separation to create a fairly dense core. Based on computer simulations first performed in 1987 by Willy Benz, Wayne Slattery, and Alastair G. W. Cameron (1925–2005), it appears that Mercury may have experienced a major collision with a large planetesimal early in its history. The collision was sufficiently energetic that much of the outer, lighter silicate material was removed, leaving behind the iron and nickel that had previously settled to the center of the planet. As a result, after the collision the planet's average density was substantially increased. Estimates place the mass of the impactor at about one-fifth Mercury's current mass, and the speed of the impact at perhaps 20 km s^{-1} . Prior to the collision, Mercury's mass may have been twice its present value. Although this may seem to be an ad hoc explanation for Mercury's unusual density, we will soon learn that the early Solar System was a violent place and that massive collisions were simply a part of its evolution.

Mercury's Weak Magnetic Field

Mercury's rotation, together with its large conducting metallic core, may be responsible for its magnetic field. The maximum strength measured by Mariner 10 was about $4 \times 10^{-7} \text{ T}$ at an altitude of 330 km, about 100 times weaker than the magnetic field measured near Earth's surface. The mechanism for generating this and other planetary magnetic fields is believed to be the magnetic dynamo, essentially the same process responsible for the Sun's magnetic field (Section 11.3). The difference between the planetary and stellar mechanisms is that a *liquid metallic* conducting core replaces the ionized gas in stars as the source of the field. To date, the details of planetary dynamos are not well understood. In Mercury's case, the fact that the rotation is so slow seems to contradict the idea that a magnetic dynamo is currently in operation. Furthermore, the relatively small size of the planet suggests that its core should have cooled to the point that any molten core would be too insignificant to generate a measurable field. As a result, opponents of a present-day dynamo mechanism suggest that Mercury's magnetic field may be a “frozen-in” remnant of its past, when the planet may have been rotating faster and when it was warmer.

20.2 ■ VENUS

Venus, the second planet from the Sun, is sometimes referred to as Earth's sister planet because its mass ($0.815 M_{\oplus}$) and radius ($0.9488 R_{\oplus}$) are comparable to Earth's. Despite these basic similarities, the two planets are markedly different in many of their fundamental features.

Retrograde Rotation

In the 1960s, one of the many unusual features of Venus was discovered. Astronomers learned that the atmospheric circulation was retrograde (the direction opposite its orbital motion), with speeds near 100 m s^{-1} at the cloud tops close to the equator; see Fig. 20.3(a). This inference was based initially on observations of the clouds in its atmosphere; it was confirmed later by measurements of the Doppler shift of spectral lines from sunlight reflected off the clouds. Later, Earth-based radar Doppler measurements of the surface (like those made of Mercury) revealed that the planet itself also rotates retrograde, but 60 times more slowly than its upper atmosphere. The sidereal rotation period of Venus is a very sluggish 243 days; this compares to its orbital period of 224.7 days.

The retrograde rotation of the planet is an interesting puzzle. All of the planets in the Solar System orbit prograde, as do most of their moons. This means that these worlds orbit in a counterclockwise direction as seen from a vantage point above Earth's north pole. Furthermore, with the exceptions of Venus, Uranus, and Pluto, all of the other planets and most of their moons also rotate prograde, just as the Sun does. This agrees with what one would expect from the development of the Solar System out of a spinning disk of material that formed when the Sun did.

Based on detailed analytical and numerical studies of the interactions among Venus, the Sun, and the other planets in the Solar System, Alexandre Correia and Jacques Laskar were able to show that Venus's retrograde rotation can be explained in terms of gravitational perturbations. The many perturbations acting on Venus from the other bodies in the Solar

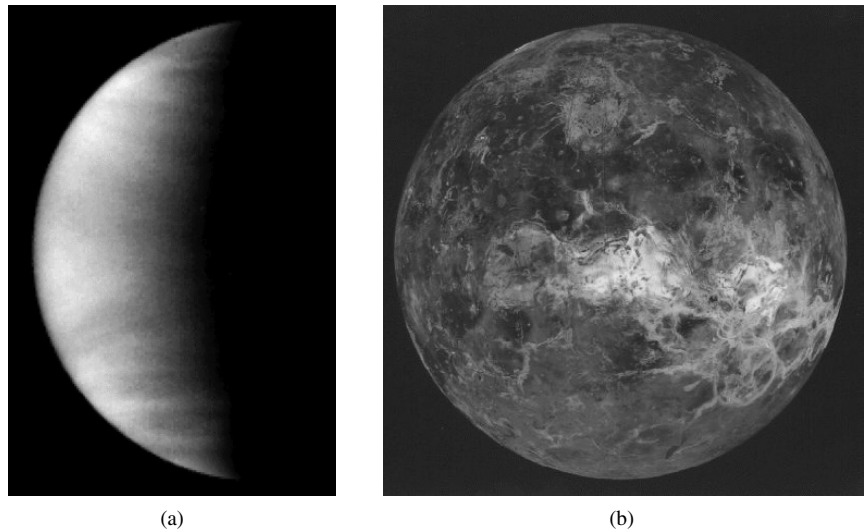


FIGURE 20.3 (a) An ultraviolet image of Venus obtained in 1995 by WF/PC 2 onboard the Hubble Space Telescope. Note the “Y”-shaped cloud features at the top of the planet’s thick atmosphere. No surface features can be seen in visible or ultraviolet light. (Courtesy of L. Esposito, University of Colorado, Boulder, and NASA.) (b) A composite radar image of the surface of Venus obtained by the Magellan spacecraft that orbited the planet from 1990 to 1994. (Courtesy of NASA/JPL.)

System lead to a chaotic zone in the tilt of the axis between 0° and 90° . As gravitational perturbations cause Venus to pass through that zone, the tilt of its rotation axis can vary dramatically. Within the first several million years following the planet's formation, its very thick atmosphere develops and also starts to influence the outcome of the planet's rotation. This occurs because the thick atmosphere can be significantly affected by tidal forces, and the thick atmosphere can also produce a damping effect. Numerical simulations based on various initial conditions for the rotation period and tilt of Venus's axis most often led to the very slow, retrograde rotation of Venus that is observed today. However, the path to this final state could be through either the flipping of the rotation axis to near 180° or the slowing of the spin rate to zero at an axis tilt of 0° , and then tides producing a slow retrograde rotation.

The dynamical behavior of Venus's atmosphere is also a puzzle. Probes that entered the atmosphere measured the presence of two large Hadley cells (Section 19.3), one in each hemisphere, consistent with the planet's slow rotation rate and the corresponding lack of any significant Coriolis force [ω is very small in Eq. (19.11)]. However, near the equator the cloud cover circles the planet in just four days, producing the "Y"-shaped cloud patterns evident in Fig. 20.3(a). Such high-speed motions are common in high-altitude jet streams (narrow rivers of air) but are unusual for the bulk of the atmosphere, particularly with such slow underlying rotation.

The Lack of a Magnetic Field

One consequence of the planet's slow rotation that does agree with expectation is the lack of any measurable magnetic field. The currents within a molten, conducting core are generated by planetary rotation; therefore, one crucial component of the magnetic dynamo mechanism is absent in Venus. Because there is no magnetic field to protect the planet via the Lorentz force (Eq. 11.2), supersonic ions in the solar wind directly strike the upper atmosphere, causing collisional ionization and a standing shock wave at the location where the solar wind particles are abruptly slowed to subsonic speeds.

The Hot, Thick Atmosphere of Venus

Analysis of the composition of the dense atmosphere, made first by ground-based telescopes and later by Soviet and American probes, revealed that its chief constituent is carbon dioxide (CO_2), which makes up about 96.5% of the total number of atoms or molecules, with molecular nitrogen (N_2) making up most of the remainder (3.5%). Traces of other molecules are also present, most notably argon (70 ppm),² sulfur dioxide (SO_2 , 60 ppm) carbon monoxide (CO, 50 ppm), and water (H_2O , 50 ppm). The probes even detected thick clouds of concentrated sulfuric acid. At the base of the atmosphere the temperature is 740 K, sufficient to melt lead, and the pressure is 90 atm, equal to the pressure at a depth of over 800 m below the surface of Earth's oceans.³

The very high surface temperature far exceeds what is expected from a simple blackbody analysis, such as the one performed in Example 19.3.1. It is the large amount of carbon

²ppm represents parts per million.

³1 atm = $1.013 \times 10^5 \text{ N m}^{-2}$.

dioxide (a greenhouse gas) in the atmosphere that is responsible for the extreme conditions at the surface. The atmosphere is so thick that the optical depth at infrared wavelengths is approximately $\tau = 70$, meaning that the temperature is increased over the blackbody temperature that would be predicted for an *airless* planet at the location of Venus by a factor of nearly $(1 + \tau)^{(1/4)} = 2.9$; see Problem 20.7.

How could Earth's sister planet have developed an atmosphere so different from our own? The formation of terrestrial atmospheres is still not well understood and is an area of active research. However, based on the direct evidence we have of outgassing from Earth's volcanoes and the discovery of volcanoes on both Venus and Mars, it seems likely that at least a portion of a terrestrial planet's atmosphere may arise from volcanic activity. It has also been suggested that significant fractions of the atmospheres of these planets may have been delivered by comets and meteorites. If the later suggestion is true, then understanding the atmospheric evolution of the terrestrial worlds requires a greater understanding of the composition of comets and meteorites, as well as the frequency with which they collide with the worlds of the inner Solar System. Comets and meteorites are discussed in more detail in Sections 22.2 and 22.4, respectively.

Whatever the source of Venus's primordial atmosphere, carbon dioxide is the dominant constituent today, and very little water is present. Conversely, water is abundant in Earth's oceans, but there is very little atmospheric carbon dioxide. What happened to change the relative abundances of those molecules on the two planets? If the two worlds began with similar compositions, as seems likely, given that they formed near one another in the solar nebula and have comparable sizes, then water was probably much more abundant on Venus in the past. In fact, since the luminosity of the zero-age main sequence Sun was only about $0.677 L_{\odot}$, much less than it is today (recall Fig. 11.1), Venus may have even had hot water oceans on its surface early in its history. As the Sun's luminosity increased and the planet was bombarded by planetesimals, the surface temperature began to rise and the oceans started to evaporate. The addition of more infrared-absorbing water vapor in the atmosphere triggered a *runaway greenhouse effect*, causing the surface temperature to climb to near 1800 K, hot enough to vaporize the remainder of the water and even melt rock. At the same time, the atmospheric pressure at the surface reached 300 atm. Since H_2O is lighter than CO_2 , the water migrated to the top of the atmosphere where it was dissociated by solar ultraviolet radiation through the reaction $H_2O + \gamma \rightarrow H + OH$. This UV photodissociation process liberated the lighter hydrogen atoms, allowing most of them to escape from the planet. Since the carbon dioxide remained, it became the dominant species in the atmosphere of Venus.

As is true of any viable scientific theory, it is important that the theory make testable predictions. In the evolutionary scenario for the atmosphere of Venus just described, the photodissociation of water should have left behind altered isotope ratios of hydrogen. Hydrogen has two stable isotopes, 1_1H (or simply hydrogen, H) and 2_1H (deuterium, D), which are chemically identical but differ in mass by a factor of two. On Earth the ratio of the numbers of deuterium atoms to hydrogen atoms is $D/H = 1.57 \times 10^{-4}$. However, within the atmosphere of Venus the ratio is closer to $D/H = 0.016$. The factor-of-100 increase in the atmospheric D/H ratio for Venus relative to Earth is due to the more massive isotope's slower rate of escape (recall Eqs. 19.8 and 19.9). Apparently our understanding of the runaway greenhouse effect on Venus is essentially correct.

Studying the Surface

Because of the thick cloud cover and inhospitable climate, it has been a difficult job gathering information about the surface of Venus. The Soviet **Venera** missions in the late 1960s through early 1980s were able to descend into the atmosphere of the planet and in some cases land on the surface, operating for short periods of time before succumbing to the environment found there. While on the surface, they returned pictures of their immediate vicinity. The landers also sampled the composition of the atmosphere and surrounding rock, confirming the presence of sulfur in the atmosphere and finding rock of volcanic origin on the surface. Variations in sulfur dioxide content over timescales of decades and the detection of radio bursts characteristic of lightning in the atmosphere support the suggestion of recent volcanic activity. In particular, observations by various spacecraft and ground-based telescopes indicate that the sulfur dioxide content in the atmosphere of Venus has decreased by more than an order of magnitude, with some intermittent fluctuations since the late 1970s. Since ultraviolet radiation converts SO_2 to sulfuric acid in the upper atmosphere, the observed decrease in the sulfur dioxide concentration has led some scientists to suggest that a major eruption may have occurred sometime during the 1970s, with a smaller event occurring around 1992.

By far the greatest amount of information about the surface of Venus has come from radar imaging, because radio signals can easily penetrate the atmosphere even though visible and ultraviolet light cannot. Radar studies have been carried out using Earth-based telescopes such as Arecibo, and from orbiters, including the **Venera** and **Pioneer** series, and most recently from the **Magellan** spacecraft. Launched from the Space Shuttle *Atlantis* in 1989, Magellan's very successful mission lasted until 1994, when it was intentionally sent diving into the atmosphere to gather information about the density structure of the planet's atmosphere. During Magellan's operational lifetime, it mapped 98% of the surface at resolutions of between 75 m and 120 m. A Magellan mosaic of one hemisphere of Venus is shown in Fig. 20.3(b).

During roughly one-half of its mission, Magellan sent back a continuous radio signal to Earth so that scientists could monitor the variation in the signal's wavelength caused by the Doppler effect. As Magellan passed over regions of higher average density, the local gravitational pull would speed up the spacecraft slightly and the wavelength of the signal received at Earth would change. In this way, Magellan was used to generate a detailed gravity map of the planet, covering approximately 95% of its surface.⁴

By combining images of a given region made from two different locations, together with gravity information, scientists have been able to produce detailed three-dimensional images of much of the planet's surface. Figure 20.4 shows Maat Mons, a volcano 8 km high located 0.9 degrees north of the equator. In this image the vertical relief has been exaggerated by a factor of 22.5 to bring out important features. Based on changes in surface reflectivity, variations in rock characteristics become evident. Apparent in the image of Maat Mons are lava flows that extend for hundreds of kilometers from the volcano. Estimates place the age of the surface rock around Maat Mons at less than 10 million years, and it may be much younger.

⁴To increase the resolution of the gravity data, flight controllers lowered Magellan's orbit using the previously untried technique of aerobraking; Magellan dipped down into the atmosphere slightly, causing the spacecraft to lose orbital energy via atmospheric drag.

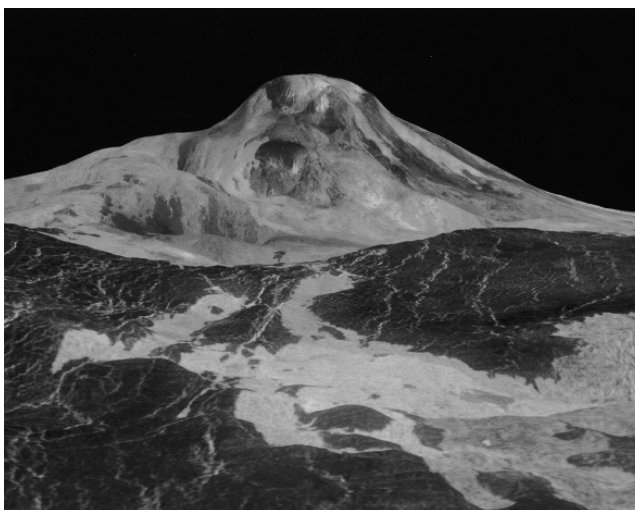


FIGURE 20.4 Maat Mons is believed to be the tallest volcano on Venus, measuring 8 km in height. The vertical scale has been increased over the horizontal scale by a factor of 22.5. (Courtesy of NASA/JPL.)

It appears that the entire surface of the planet may have been refreshed relatively recently compared to the age of the Solar System. This estimate comes from the number of impact craters found on the surface (e.g., see Fig. 20.5). If we assume that Venus has been struck by impacts at about the same rate as other worlds in the inner Solar System (such as Mercury or our Moon), then from the relatively low number of craters found on the surface of Venus we can conclude that large-scale lava flows must have occurred about 500 million years ago.⁵ In support of this conclusion, nearly one thousand volcanic features have been identified on the surface of the planet.

20.3 ■ EARTH

The planet for which we have the greatest amount of information by far is of course our Earth (see Fig. 20.6). We have studied its atmosphere, its oceans, and its active geology with a great deal of specificity. We have been able to carefully investigate its extensive biology, from the smallest microbes to the largest plants and animals, and to study the evolutionary processes that have led to our planet's great biodiversity. We have also been able to extend our knowledge by developing follow-up experiments based on information gained from previous studies. This makes investigating our own planet significantly more robust and interactive than investigating other bodies in our Solar System to date.⁶

⁵Absolute age estimates of the Moon will be discussed in detail in Section 20.4.

⁶However, as we will see when we discuss Mars in Section 20.5, humans have begun to conduct extensive robotic studies of that planet based directly on information returned from previous and ongoing missions.

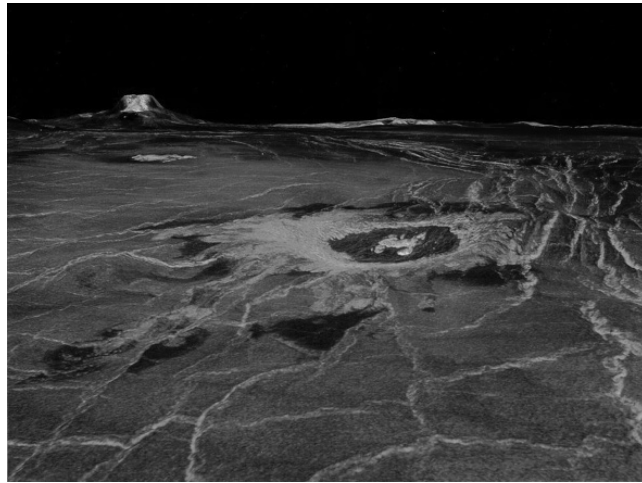
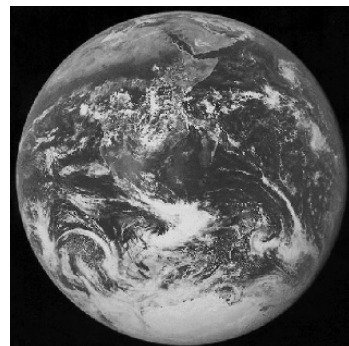


FIGURE 20.5 The impact crater Cunitz is visible in this image, with the volcano Gula Mons in the background. The vertical scale has been increased over the horizontal scale by a factor of 22.5. (Courtesy of NASA/JPL.)



(a)



(b)

FIGURE 20.6 (a) Earthrise over the limb of the Moon on December 22, 1968. This picture was taken by the astronauts of *Apollo 8*. (b) Earth as seen by the *Apollo 17* astronauts while traveling to the Moon on December 7, 1972. Visible are most of Africa, Saudi Arabia, and the south polar ice cap. (Courtesy of NASA.)

Our Atmosphere

Beginning early in its history, the bulk of Earth's water condensed to form its oceans. Unlike Venus, however, given Earth's slightly greater distance from the Sun, our planet never got hot enough to turn much of the liquid to vapor (see Eq. 19.5). Therefore, the ensuing runaway greenhouse effect described in Section 20.2 never developed. Instead, the carbon dioxide in the atmosphere was dissolved into water, where it became chemically bound up in carbonate rocks such as limestone. If all of the carbon dioxide trapped within

rock today were released into Earth's atmosphere, the amount would be comparable to that currently contained in the atmosphere of Venus.

However, it is also important to recall from Fig. 11.1 that the Sun was significantly less luminous in the early Solar System than it is today. This implies that Earth's surface would have been cooler in the past and its water should have been in the form of ice, even as recently as 2 billion years ago. However, geologic evidence, including fossil records, suggests that Earth's oceans were liquid as early as 3.8 billion years ago. This puzzle has become known as the *faint ancient Sun paradox*. The resolution of this paradox probably lies in details of the greenhouse effect and a different atmospheric composition than exists at the present time.

The present-day atmosphere of Earth is made up of (by number) 78% N₂, 21% O₂, 1% H₂O, and traces of Ar, CO₂, and other constituents. The atmosphere owes its current composition in part to the development of life on the planet. For instance, plants process carbon dioxide into oxygen as a by-product of photosynthesis.

The Greenhouse Effect and Global Warming

Serious concern now exists over the effects of artificially introducing carbon dioxide and other greenhouse gases into Earth's atmosphere by industrial means. To complicate matters, we are simultaneously destroying vast regions of vegetation, such as the Amazon rain forests, that could recycle CO₂. The commonly used technique of slash-and-burn clearing of the rain forests also releases tremendous quantities of carbon dioxide into the atmosphere.

To illustrate the problem, recent changes in the abundance of carbon dioxide over Mauna Loa, Hawaii, are shown in Fig. 20.7; the oscillations are due to the annual growing season.

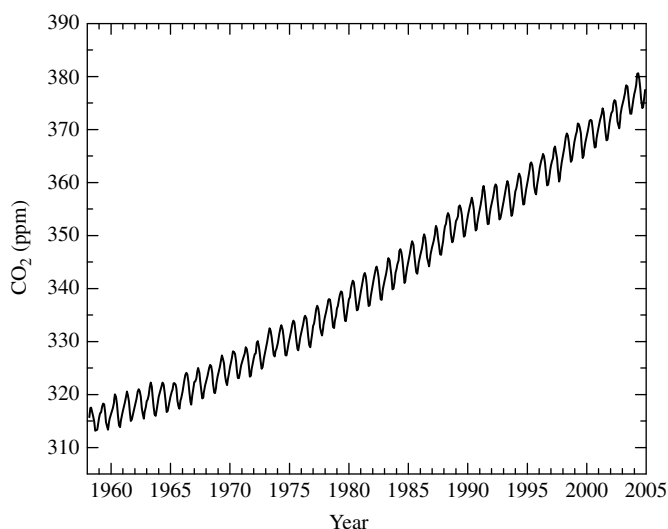


FIGURE 20.7 The amount of carbon dioxide in parts per million by volume (ppm) over Mauna Loa, Hawaii, as a function of time. (Data from C. D. Keeling, T. P. Whorf, and the Carbon Dioxide Research Group, Scripps Institution of Oceanography, University of California.)

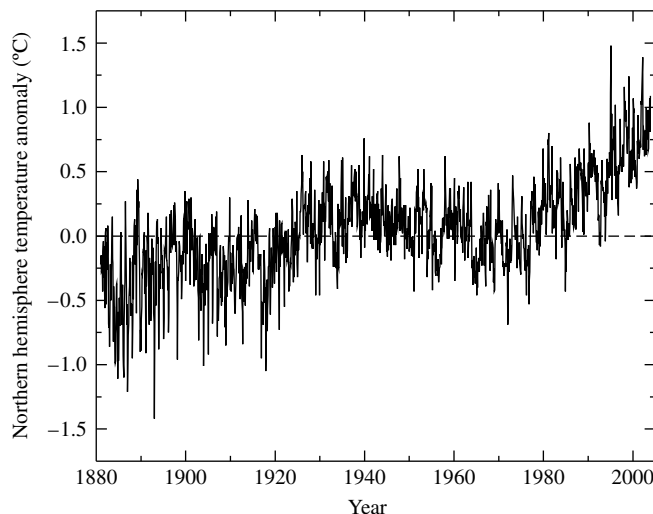


FIGURE 20.8 Monthly average temperature deviations in the Northern Hemisphere of Earth from 1881 to 2003. The deviations are measured from a 25-year average between 1951 and 1975. (Data from K.M. Lugina, P.Ya. Groisman, K.Ya. Vinnikov, V.V. Koknaeva, and N.A. Speranskaya, 2004. *In Trends Online: A Compendium of Data on Global Change*. Carbon Dioxide Information Analysis Center, Oak Ridge National Laboratory, U.S. Department of Energy, Oak Ridge, Tennessee, U.S.A.)

Because of the nonlinear behavior of the greenhouse effect and the very complicated physics, chemistry, and meteorology that are involved, accurate computer models are only slowly becoming available. However, despite current limitations in the predictive power of these models, the basic effects of greenhouse gases are understood. As we have learned from Venus, increasing the content of greenhouse gases in an atmosphere will raise its average temperature. The questions are by how much the temperature will increase, and how rapidly it will occur.

Figure 20.8 shows the average temperature deviations in the Northern Hemisphere of Earth between 1881 and 2003 with respect to a 25-yr average of temperatures computed from 1951 through 1975. Evident is a consistently upward trend in the average temperature since about 1970. Whether this upward trend is the start of a long-term steady increase or a fairly short-term fluctuation has been a matter of some debate. However, it is clear that a significant upward trend is currently under way; in fact, seven of the ten hottest years in the twentieth century occurred in the 1990s.

Associated with the effects of **global warming** is evidence that Earth's glaciers are receding world wide. In addition, the Arctic ice cap has thinned significantly since 1970 and Earth's ocean levels have risen. Also supporting the conclusion that human-driven global warming is occurring is the increase in the average ocean surface temperature by approximately 0.5°C since the late 1960s, with warming extending down to depths of several hundred meters. Since the oceans eventually absorb some 84% of the excess heat in the atmosphere, the observation of this temperature increase is significant. The ocean temperature increase also agrees with computer modeling of global climate changes that include the influence of increased greenhouse gas emissions.

Another environmental concern about human activity is the release of chlorofluorocarbons into the atmosphere. These molecules migrate into the upper atmosphere above the North and South Poles, where they are destroying the ozone (O_3). Ozone is known to be a major absorber of ultraviolet radiation, and as such, it plays an important role in protecting life on Earth's surface.

Much more research is necessary before we can hope to understand the magnitude of the environmental consequences of human behavior. Unfortunately, by the time more detailed predictions are available, it may not be possible to reverse the trend.

In recognition of the importance of global warming to the inhabitants of our planet, the first-ever "Earth Summit" was held in 1992, involving most of the nations of the world. Officially known as the United Nations Conference on Environment and Development, its purpose was to discuss global environmental concerns. The treaty that came from that summit is the Framework Convention on Climate Change. Then, in December 1997, more than 160 nations met in Kyoto, Japan, to negotiate binding limitations on greenhouse gases for the developed nations. After much debate and compromise, resolutions regarding such things as the emissions of greenhouse gases were finally agreed on. The Kyoto Protocol came into force on February 16, 2005, after being ratified by 157 countries. However, the United States, the world's largest producer of greenhouse gases at the time the Kyoto Protocol went into effect, did not ratify the agreement, citing concerns over its impact on the nation's economy.

Seismology and Earth's Interior

The structure of Earth's interior can be derived by analyzing the seismic waves generated by earthquakes. Two principal types of waves are produced by earthquakes: **P waves** (pressure or *primary*) are longitudinal waves capable of traveling through both liquids and solids, and **S waves** (shear or *secondary*) are transverse waves that are restricted to traveling through solids only (see Fig. 20.9). Since the velocities and paths of both P and S waves depend on the medium through which they are moving, their detection around the world enables geologists to deduce the structure of our planet.⁷ For instance, in regions where only P waves are measured, the absence of S waves implies that there must have been intervening liquid in the path of the wave (see Fig. 20.10). Furthermore, because of the refraction that occurs at boundary interfaces (much like the refraction of light rays at the boundaries between media of differing indices of refraction), **shadow zones** exist where neither type of wave can be detected. Thus, using the data from P and S waves, geologists can map the interior of the planet. Such maps yield information about the depth of the surface **crust** and reveal the existence of a solid **inner core**, a molten **outer core**, and a thick **mantle**.

The behavior of P waves in the outer core implies that its composition is predominantly iron and nickel. This also agrees with the fact that the average density of Earth is 5515 kg m^{-3} , greater than the density of surface rocks (typically 3000 kg m^{-3}) and water (1000 kg m^{-3}).⁸ It is the combination of high temperature ($> 4000 \text{ K}$) and composition that results in a *liquid outer core*. The transition back to a *solid inner core* occurs because of the extreme pressures found there.

⁷An analogous procedure is used to study the interiors of many stars; recall Chapter 14.

⁸Gravitational compression also contributes to the higher value for the average density, relative to surface material.

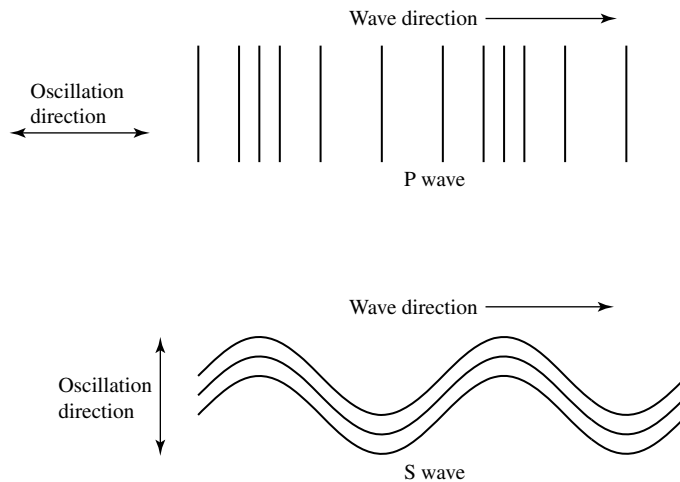


FIGURE 20.9 P waves are longitudinal pressure waves capable of traveling through both liquids and solids. S waves are transverse shear waves that can travel only through solids.

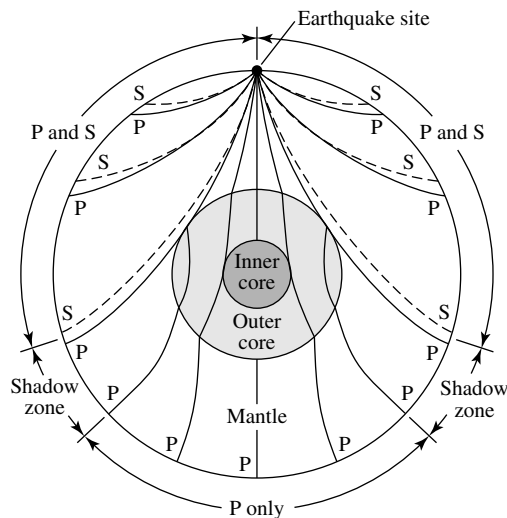


FIGURE 20.10 Generated by earthquakes, P and S waves travel through Earth's interior. S waves are unable to traverse the molten outer core. Furthermore, the refraction of P waves at the interface between the outer core and the mantle produces shadow zones.

Plate Tectonics

Although the presence of volcanos is a feature that Earth shares with Venus and Mars, Earth's present-day **tectonic activity** appears to be unique among the terrestrial planets. This activity has its origin in the dynamic interior of Earth, depicted in Fig. 20.11. Earth's surface layer, known as the **lithosphere**, encompasses both the oceanic and continental crust as well as the outer portion of the mantle. The lithosphere is fractured into **crustal plates**

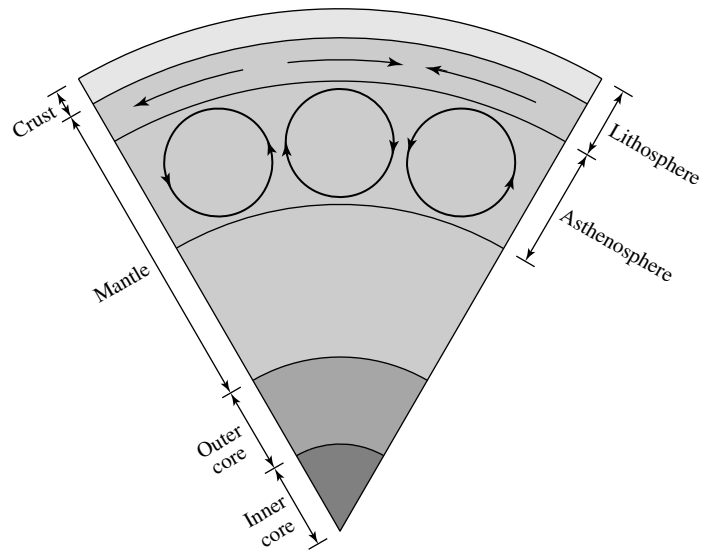


FIGURE 20.11 The interior structure of Earth is composed of an inner solid core, an outer liquid core, a mantle, and surface crust. The crust and the outer portion of the mantle make up the lithosphere (containing the surface plates) and the underlying convective asthenosphere. The diagram is not to scale.

(see Fig. 20.12) and rides on the convective, somewhat plastic **asthenosphere**, which is also part of the mantle. As the plates move across the surface of the planet, they crash into or grind against one another, carrying the continents with them.⁹ Because of these motions, the Atlantic Ocean is widening at the rate of approximately 3 cm yr^{-1} , spreading away from an underwater mountain range that runs the length of the ocean floor. This mid-Atlantic ridge is the location where material from the interior rises to the surface, generating new sea floor as the continents separate (see Fig. 20.13).

Extrapolating the motions of the plates backward in time, geologists believe that there was once one giant supercontinent, known as Pangaea, that broke apart some 200 million years ago into two smaller supercontinents, Laurasia and Gondwanaland. Gondwanaland in turn separated into South America and Africa, and Laurasia divided into Eurasia and North America.

Earth's plate boundaries are generally the sites of active volcanism, mountain building, and frequent earthquakes. For example, when two plates collide, the lighter continental crust overrides the heavier oceanic crust and a **subduction zone** develops, as illustrated in Fig. 20.13. One such location is along the coast of Japan, where its volcanic islands were created as a result of the heat generated by friction as the oceanic crust descended into the interior of Earth. It is at the location of these subduction zones that deep oceanic trenches also develop. If two plates collide that contain continental crust, neither plate will overrun the other; instead, buckling occurs and a mountain range such as the Himalayas is generated.

⁹For example, the Pacific and North American plates are currently sliding past one another. The famous San Andreas fault is located on the boundary between these two plates.

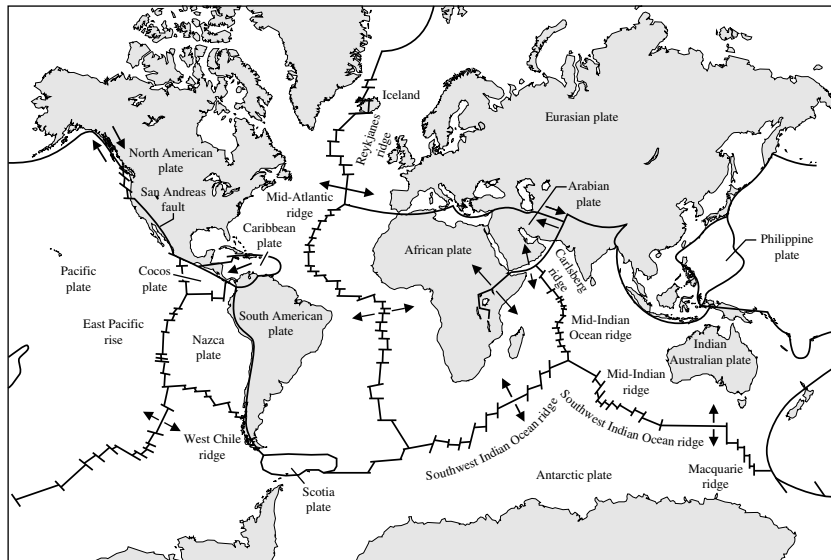


FIGURE 20.12 The lithosphere is divided into crustal plates that travel across Earth's surface.

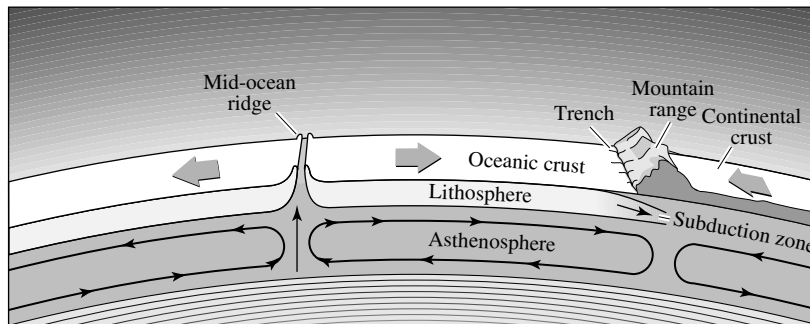


FIGURE 20.13 The motions of the plates are driven by convection zones in the asthenosphere. A mid-ocean ridge (a rift) occurs where material from below is pushed up to the surface. A subduction zone develops when the lighter continental crust overrides the heavier oceanic crust of two colliding plates.

Sources of Internal Heat

All of this activity requires one or more sources of energy to sustain itself. Heat is known to be escaping into space through the surface of Earth at a rate of $4 \times 10^{13} \text{ W}$, implying an average flux of 0.078 W m^{-2} . If the only source of energy in the interior were the heat left over from the formation of the planet almost 4.6 billion years ago, plate tectonic activity would have long since ceased. Other sources of heat augment the *energy budget* of Earth, including the tidal dissipation of its rotational kinetic energy (see Problem 20.10), possible ongoing gravitational separation (releasing gravitational potential energy as heavier

constituents sink toward the center of the planet), and the continual radioactive decay of unstable isotopes (believed to be the primary source of the heat).¹⁰ This allows much of the interior to remain somewhat plastic, supporting the large, sluggish convection cells that drive the motions of the crustal plates.

Earth's Variable Magnetic Field

The presence of a molten iron–nickel outer core together with Earth's relatively rapid rotation rate is consistent with the observation that the planet possesses a global magnetic field, assuming that a dynamo is in operation in the planet's interior. The existence of Earth's magnetic field serves to protect the planet from incoming charged particles in the solar wind, as well as other ionized cosmic rays. Instead of striking the surface, these particles become trapped in the dipole field and bounce back and forth between the North and South poles (Fig. 20.14). Three regions of trapped particles have been identified and are known as the **Van Allen radiation belts**. The innermost belt is composed of protons and is at a height of roughly 4000 km above Earth's surface. Overlapping a portion of the inner belt is a second belt composed of atomic nuclei that were once part of the interstellar medium. The outermost belt is composed of electrons at an altitude of approximately 16,000 km. Particles in the belts that are energetic enough to enter Earth's upper atmosphere near the poles strike atoms and molecules there, causing collisional excitation, ionization, and dissociation. When the atoms or molecules recombine, or when the electrons drop back down to lower energy levels, the subsequent emission of light is observed as the **aurora borealis** (northern lights) and **aurora australis** (southern lights); see Fig. 20.15.

Interestingly, geologic evidence indicates that Earth's magnetic field weakens, reverses polarity, and reestablishes itself on an irregular time scale of some 10^5 years. This can be seen in the orientation of magnetic minerals trapped in molten rock that later solidified,

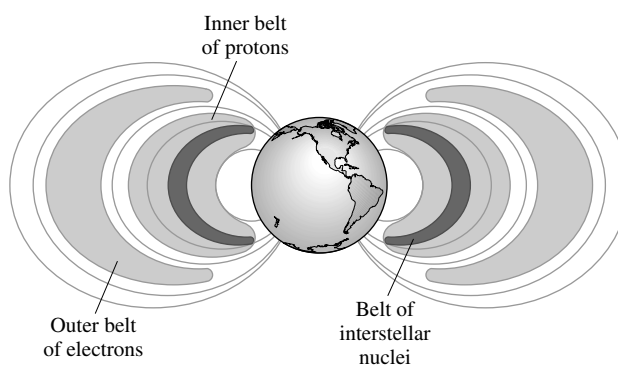


FIGURE 20.14 The Van Allen radiation belts arise from charged particles becoming trapped in the magnetic field of Earth.

¹⁰In the mid-1800s, Lord Kelvin argued that Earth could not be more than about 80 million years old. His argument was based on the amount of gravitational potential energy the planet could release and the rate at which heat escaped over time. However, his calculation was made before the discovery of radioactivity.



FIGURE 20.15 The aurora are due to the collision of high-speed particles with atoms and molecules in Earth's upper atmosphere. (Courtesy of the Geophysical Institute, University of Alaska, Fairbanks.)

such as those sampled on either side of the spreading mid-Atlantic ridge (Figs. 20.12 and 20.13). In this way, a “fossil record” of the direction of the local magnetic field is created. The behavior of Earth's field is not unlike the solar cycle, the flipping of the Sun's magnetic field roughly every 11 years (Section 11.3). Earth's magnetic field is known to be weakening today.

20.4 ■ THE MOON

Despite the proximity of the Moon to Earth, the two worlds are very different (see Fig. 20.16). Because of its low surface gravity, the Moon has been unable to retain a significant atmosphere. Without a protective atmosphere, the Moon has suffered impacts by meteorites throughout its history. Along with a large number of smaller impacts, a significant number of very large collisions occurred approximately 700 million years after the Moon formed. These impacts were powerful enough to penetrate its thin crust, allowing molten rock in the interior to flow across the surface. The result was the formation of the many smooth, roughly circular **maria** (or “seas”) that can be seen on the surface of the Moon facing Earth. It is the distribution of these maria that has led humans to imagine seeing the face of the “man in the Moon.”

The Moon's Internal Structure

Major advances in our understanding of the Moon's internal structure and evolutionary history have occurred as a result of intense exploration from 1959 through the early 1970s. When the **Apollo** astronauts landed on the Moon, they left seismic detectors designed to measure any *moonquakes* that may be occurring. Many of the very weak quakes that were detected (about magnitude 1 on the Richter scale) were triggered by the tidal strain generated by Earth's gravitational pull. Another class of vibrations has been attributed to the Moon's

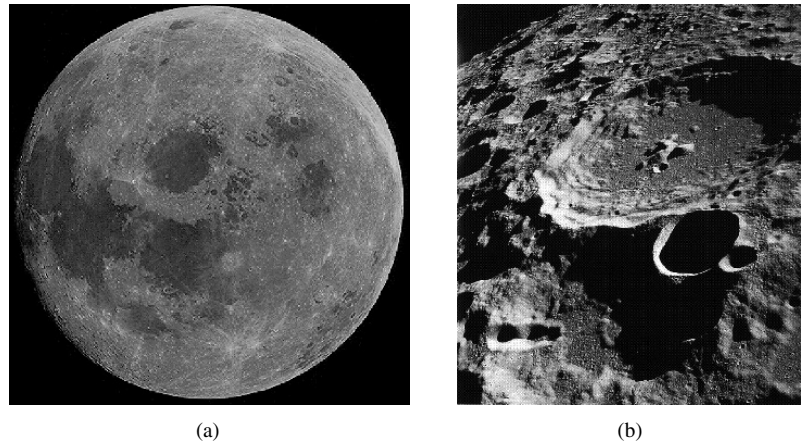


FIGURE 20.16 (a) The surface of the Moon contains heavily cratered highland regions and nearly circular maria that are much less heavily cratered. The portion of the Moon facing Earth is on the left side of the image. (b) A portion of the far side of the Moon showing the extensive cratering found there. The diameter of the large crater is approximately 80 km. This view was obtained by the *Apollo 11* astronauts in 1969. (Courtesy of NASA.)

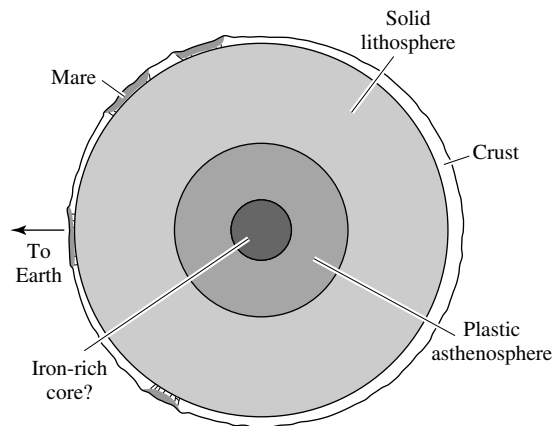


FIGURE 20.17 The interior structure of the Moon.

“ringing” after being struck by meteorites. Just as with the analysis of seismic activity on Earth, moonquakes have allowed scientists to develop an understanding of the Moon’s interior.

Many of the moonquakes appear to have originated, not at tectonic plate boundaries, but at the interface between the solid, brittle lithosphere and the plastic asthenosphere (see Fig. 20.17). It also appears likely that below the asthenosphere, a small iron-rich core exists as well. This structure is consistent with measurements of the small amount of heat still flowing outward from the Moon’s interior, which is responsible for maintaining the plastic

nature of the asthenosphere. However, based on the data provided by this seismic activity, Earth-like tectonic activity appears to be absent on the Moon today.

Interestingly, only one mare was seen on the side of the Moon farthest from Earth.¹¹ This is not because collisions were preferentially occurring on the side of the Moon facing Earth; rather, the crust is actually thinner on the near side. Consequently, impacts on the thin-crust side were more likely to penetrate the crust, allowing interior molten rock to flow over the surface. Because the crust is less dense than material in the Moon's interior, tidal forces have caused the heavier near side to permanently "hang down" toward Earth.

The Absence of a Global Magnetic Field

Unlike Earth, our Moon has no measurable global magnetic field, apparently because the Moon is small enough to have cooled off much more rapidly than Earth. This evolution has left the Moon as a geologically inactive world today. Furthermore, the Moon's rotation period is more than 27 times longer than Earth's. As a result, there is no evidence of any significant magnetic dynamo in operation, suggesting that if a molten core is present, it is likely to be quite small.¹²

The lack of a global magnetic field on the Moon makes the detection of a weak field around Mercury even more puzzling (recall the discussion on page 740). The two worlds are comparable in mass and radius (the Moon's mass and radius are 23% and 71% of the values for Mercury, respectively), while Mercury's rotation rate is slower by a factor of two. Clearly, much work remains to be done in understanding the details of magnetic field production.

Moon Rocks

During the 1960s and 1970s, six manned United States Apollo missions returned 382 kg of surface rocks and regolith from the Moon's surface. In addition, three unmanned Soviet Union **Luna** missions returned an additional 0.3 kg of material. The samples were collected from both the maria and the **highland** (or mountainous) regions between the maria. These samples represent the most detailed information we have about the nature of our closest neighbor.

Composition analysis of samples returned from the maria confirm that they are in fact volcanic in origin. The rocks are **basalts**, similar to the kind of volcanic rock found on Earth. The lunar basalts are rich in iron and magnesium, and they also contain glassy structures that are characteristic of rapid cooling. However, unlike Earth basalts, the lunar samples contain no water and a lower percentage of **volatiles** (elements or compounds with low melting and boiling temperatures) relative to **refractories** (higher melting and boiling temperatures).

Radioactive Dating

Perhaps the most eagerly awaited results of the analysis of the lunar samples were the determinations of their ages. The process is based on measuring the abundances of certain

¹¹The first observations of the far side of the Moon were made by the Soviet Luna 3 mission in 1959.

¹²Based on the natural remnant magnetization of returned lunar samples and the patchy magnetization detected by satellites, it appears that the Moon once had a global magnetic field. However, there is no evidence of a global field today.

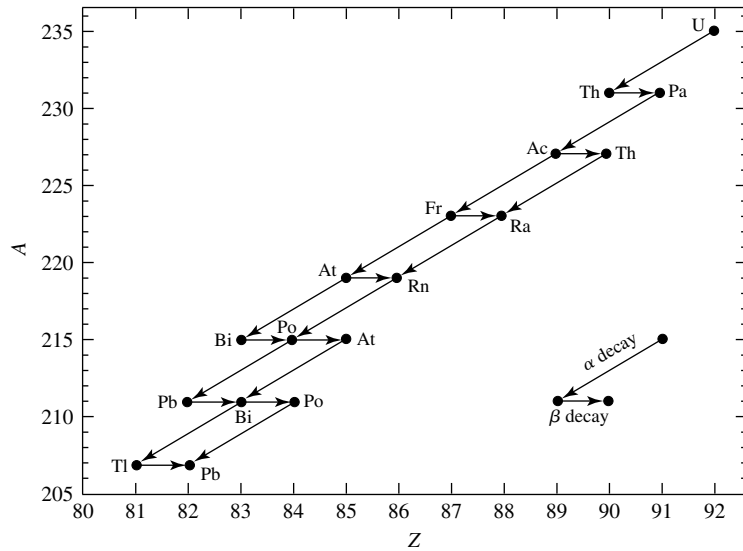


FIGURE 20.18 The $^{235}_{92}\text{U}$ decay sequence.

radioactive isotopes and comparing them with the abundances of stable end products of the decay sequence. In this technique of **radioactive dating**, we assume that the “clock” started ticking when the rock solidified, trapping the isotopes inside.

If the half-life of one step in the decay sequence is significantly longer than any of the others, it can be assumed that the original isotope decays directly into the final product with a half-life approximately equal to that of the longest one. For instance, in the decay sequence depicted in Fig. 20.18, which begins with $^{235}_{92}\text{U}$ and ends with $^{207}_{82}\text{Pb}$, the first step, the alpha particle¹³ decay $^{235}_{92}\text{U} \rightarrow ^{231}_{90}\text{Th} + ^4_2\text{He}$, has a half-life of 7.04×10^8 years, while the next slowest step, $^{231}_{91}\text{Pa} \rightarrow ^{227}_{89}\text{Ac} + ^4_2\text{He}$, has a half-life of only 3.276×10^4 years. As a result, to a good approximation, the half-life of the entire sequence can be taken to be 7.04×10^8 years. This means that by measuring the relative abundances of the uranium and lead isotopes, we can determine the time required for the transformation.

Some radioactive isotopes that are useful for dating Moon rocks, as well as Earth rocks and meteorites, are given in Table 20.1. Note that the stable products are not necessarily the direct result of a single decay but may be produced after a succession of decays, the longest of which has the quoted half-life.

To understand the method of radioactive dating more fully, suppose that isotope A decays into isotope B (which is stable), either directly or indirectly through a series of steps. From Eq. (15.10) we know that if the number of atoms of A in the sample was initially $N_{A,i}$, then after some time t , the number remaining is

$$N_{A,f} = N_{A,i}e^{-\lambda t},$$

¹³Recall that helium nuclei (^4_2He) are often referred to as alpha particles (α).

TABLE 20.1 Radioactive Isotopes with a Half-Life Useful for Determining Geologic Ages.

Radioactive Parent	Stable Product	Half-Life (10 ⁹ yr)
¹²⁹ ₅₃ I	¹²⁹ ₅₄ Xe	0.016
²³⁵ ₉₂ U	²⁰⁷ ₈₂ Pb	0.704
⁴⁰ ₁₉ K	⁴⁰ ₁₈ Ar	1.280
²³⁸ ₉₂ U	²⁰⁶ ₈₂ Pb	4.468
²³² ₉₀ Th	²⁰⁸ ₈₂ Pb	14.01
¹⁷⁶ ₇₁ Lu	¹⁷⁶ ₇₂ Hf	37.8
⁸⁷ ₃₇ Rb	⁸⁷ ₃₈ Sr	47.5
¹⁴⁷ ₆₂ Sm	¹⁴³ ₆₀ Nd	106.0

where

$$\lambda = \frac{\ln 2}{\tau_{1/2}}$$

is the decay constant and $\tau_{1/2}$ is the half-life. Because the total number of atoms of A and B must remain constant over time (even though A is ultimately being converted into B), it is necessary that

$$N_{A,f} + N_{B,f} = N_{A,i} + N_{B,i}.$$

Solving for $N_{A,i}$, substituting into the decay equation, and rearranging, we have an expression for the *change* in the number of atoms of B within the sample since it formed:

$$N_B - N_{B,i} = (e^{\lambda t} - 1) N_A,$$

where $N_A \equiv N_{A,f}$ and $N_B \equiv N_{B,f}$ are the numbers of atoms of species A and B respectively, remaining today. When comparing one sample with another, it is more accurate to evaluate the compositions by using ratios of isotopes: the isotopes of interest relative to a stable third isotope. Representing this third (constant) abundance as N_C , we arrive at the relation

$$\frac{N_B}{N_C} = (e^{\lambda t} - 1) \frac{N_A}{N_C} + \frac{N_{B,i}}{N_C}. \quad (20.1)$$

Equation (20.1) is used to determine the age of a sample by plotting relative abundances of the stable product versus the relative abundances of the radioactive isotope in the sequence at various locations in the rock. The slope $m = e^{\lambda t} - 1$ of the best-fit line is directly related to the age of the sample.

Example 20.4.1. Data for one sample obtained in the lunar highlands, based on the beta decay¹⁴ of rubidium-87 to strontium-87, ${}^{87}_{37}\text{Rb} \rightarrow {}^{87}_{38}\text{Sr} + e^- + \bar{\nu}$, are shown in Fig. 20.19. From Eq. (20.1) and Fig. 20.19,

$$m = e^{\lambda t} - 1 = 0.0662,$$

where $\lambda = 0.0146 \times 10^{-9} \text{ yr}^{-1}$ for ${}^{87}_{37}\text{Rb}$. Solving for t , we find that the age of the sample is $4.39 \times 10^9 \text{ yr}$.

It is important to point out that this procedure assumes that the initial ratio ${}^{87}_{38}\text{Sr}/{}^{86}_{38}\text{Sr}$ is a constant throughout the sample, whereas the initial ratio ${}^{87}_{37}\text{Rb}/{}^{86}_{38}\text{Sr}$ may vary somewhat (i.e., the sample is not perfectly homogeneous). This is because ${}^{86}_{38}\text{Sr}$ and ${}^{87}_{38}\text{Sr}$ are chemically identical, allowing them to be bound up in minerals in the same proportions, whereas the proportion of ${}^{87}_{37}\text{Rb}/{}^{86}_{38}\text{Sr}$ need not be constant throughout.

The results of radioactive dating are consistent with the view that the lunar maria have relatively young surfaces. As you will demonstrate in Problem 20.14 for one sample returned from the Sea of Tranquility by the *Apollo 11* astronauts in 1969, the ages of the maria (typically 3.1 to 3.8×10^9 years) are significantly less than those of the highlands. This is consistent with the observation, noted earlier, that relatively few craters can be found in the maria compared to the highland regions.

In sharp contrast, the oldest rocks ever found on Earth date to 3.8 billion years, whereas 90% of the planet's crust is younger than 600 million years. Plate tectonic activity is constantly recycling the surface, carrying old crust down into the mantle and forming new crust to replace it.

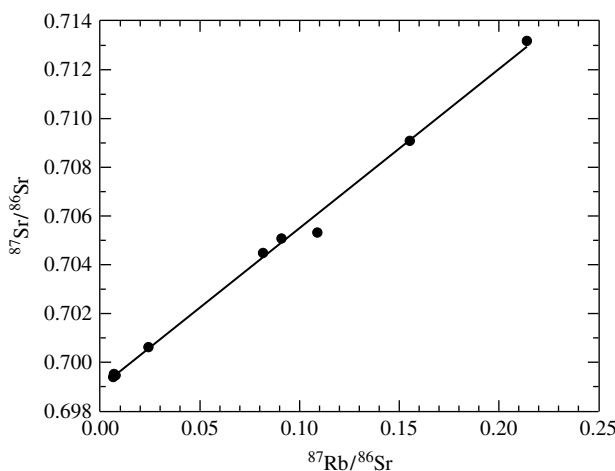


FIGURE 20.19 Relative abundance determinations for a sample obtained in the lunar highlands. (Data from D. A. Papanastassiou and G. J. Wasserburg, *Proc. Seventh Lunar Sci. Conf.*, Pergamon Press, New York, 1976.)

¹⁴Recall that an electron is also referred to as a beta particle (β).

Late Heavy Bombardment

It is the dating of lunar samples that implies that a spike of **late heavy bombardment** (LHB) occurred roughly 700 million years after the Moon formed. It was during that time that the majority of the cratering occurred in the lunar highlands. During the LHB phase, a small number of very large collisions produced the maria. Over the last 3.8 billion years meteorite impacts have continued, but at a significantly reduced rate. In this way the fairly smooth, relatively uncratered surfaces of the maria have been maintained.

The “time stamp” provided by Moon rocks not only plays an important role in our understanding of the evolution of the Moon but is also crucial in developing a picture of evolution for other planets, as well as an overall formation theory for the Solar System (see Section 23.2). For instance, the recognition of an LHB episode followed by meteorite impacts at roughly a constant rate has helped scientists to conclude that the surface of Venus was refreshed within approximately the last 500 million years. This scenario also suggests that the surface of Mercury is, in general, quite ancient.

The Formation of the Moon

The question of the Moon’s formation has been widely debated. Prior to the Apollo and Luna missions, several models had been proposed. The **fission model** (also sometimes called the **daughter model**), first suggested in 1880 by George Darwin¹⁵ (1845–1912), contended that the Moon was “torn off” from Earth at a time when Earth was spinning more rapidly than it is today. However, the orientation of the Moon’s orbital plane is close to the ecliptic (tilted 5.1°), rather than along the plane of Earth’s equator as would be expected if the Moon broke away. Furthermore, the lack of any water in the lunar samples, together with the underabundance of other volatiles relative to surface rock on Earth, also contradicts this proposal.

The **co-creation model** (also known as the **sister model**) suggested that the Moon and Earth formed simultaneously, with the Moon coalescing from a small disk of material that developed around the proto-Earth. This idea also fails to explain the composition differences found in the lunar samples.

A third model, the **capture model**, proposed that the Moon was actually formed elsewhere in the solar nebula and was caught in Earth’s gravitational field as it drifted by. However, in this scenario the composition differences are not great enough; the Moon and Earth are too similar. For instance, the ratios of stable isotopes of oxygen are nearly identical within lunar and terrestrial samples despite significant differences found in meteorites. Also, the dynamics of such a capture seem unlikely. Since the Moon is fairly large compared to Earth, a third, similarly sized body would need to have been present to take up much of the system’s surplus energy, as required for a capture. Having three large objects in close proximity at just the right time seems highly improbable. On the other hand, capture seems to be a likely mechanism for some of the many small moons found throughout the Solar System. In these cases, energy may have been lost through a many-body interaction with other moons already present. Alternatively, orbital energy may have been lost by aerobraking if the captured moon passed through a portion of the planet’s atmosphere, much like the

¹⁵George Darwin was the son of Charles Darwin (1809–1882), the author of the theory of Darwinian biological evolution.

maneuver performed with the Magellan spacecraft around Venus. However, our Moon is much too large for any of these mechanisms to work.

In 1975 a fourth model, now known as the **collision model**, was proposed by William K. Hartmann and Don R. Davis. Since that time numerous computer simulations have verified its plausibility (e.g., Fig. 20.20). This model seems to explain many of the problems encountered by the three previous scenarios. The model suggests that a giant object, perhaps

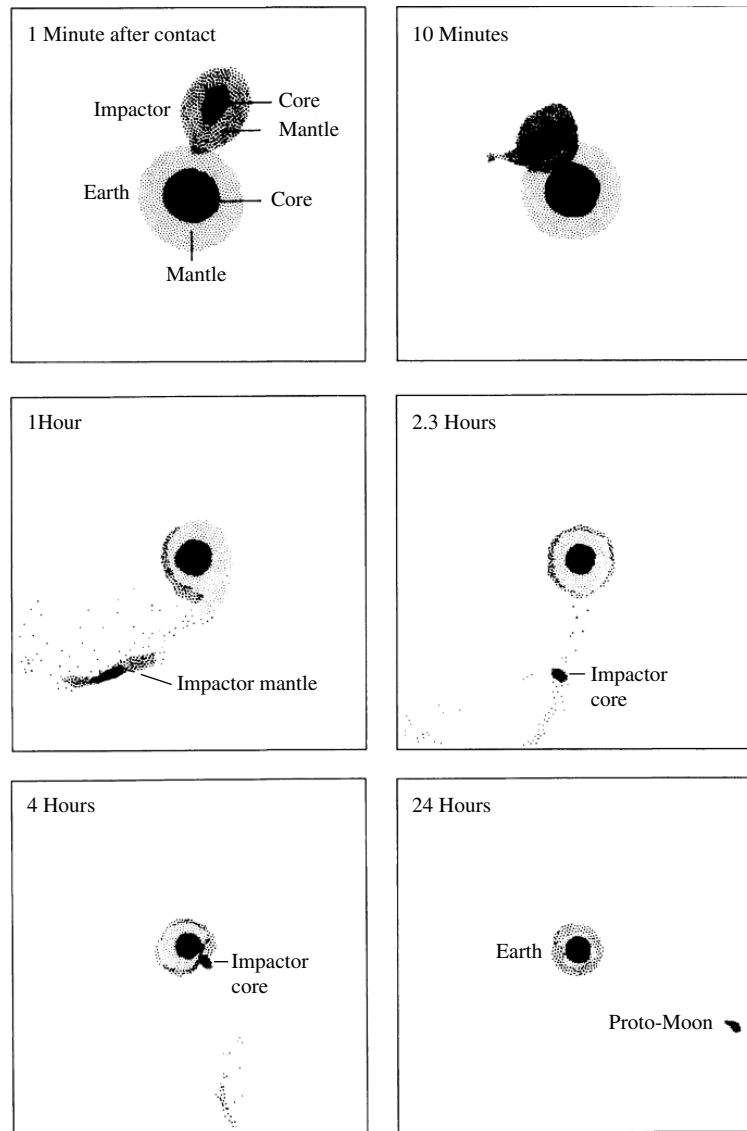


FIGURE 20.20 A computer simulation of the formation of the Moon according to the collision model. The Earth–Moon system is shown at different times during the simulation. (Figure courtesy of A. G. W. Cameron and W. Benz, Smithsonian Astrophysical Observatory.)

twice the mass of the present-day Mars, collided with Earth almost 4.6 billion years ago, vaporizing much of the impactor and causing a portion of Earth's surface to be ripped away. The pulverized material then formed a disk around Earth that coalesced over a relatively short period of time (estimates range from months to about 100 years). Due to the high temperatures generated during the collision, many of the volatiles present in Earth's crust would have been absent in the condensing debris. Assuming that there had been sufficient time prior to the collision for some gravitational separation to occur within Earth and the impactor, the crusts of the two objects would also have been somewhat deficient in iron, leaving less iron available to form the Moon. The simulations suggest that most of the present-day Moon was produced by silicate-rich material from the mantle of the impactor, and that the impactor's iron-rich core became a part of Earth. This model effectively explains why the Moon's average density is comparable to the density of Earth's uncompressed mantle (i.e., the density that would be measured if the compressional effect of gravity were removed). In this model the collision would have also preserved the similar oxygen-isotope ratios seen on the two worlds.

The collision model is considered by most researchers as the preferred model for the formation of our Moon. Although it appears on first inspection to be a highly unique and perhaps ad hoc way to explain the characteristics of our Moon, recall that a similar scenario also appears to explain the highly dense structure of Mercury (page 740). The existence of Pluto's moon Charon may require a large-scale collision as well (see Section 22.1).

From our investigation of the Moon, it appears that its formation was a violent process. However, numerous questions about the Moon's structure and evolution remain unanswered. It is also apparent that careful studies of our nearest neighbor can shed light on important questions regarding the formation and evolution of Earth and the rest of the Solar System. Perhaps future missions to the Moon would further clarify our understanding of the Solar System.

20.5 ■ MARS

Only one-tenth the mass of Earth, the planet Mars has touched our imagination. In 1877 the astronomer Giovanni Virginio Schiaparelli (1835–1910) reported seeing a series of dark lines on the surface of the planet and referred to them as *canali* (naturally occurring channels of water). The term was later misinterpreted to imply that the markings were actually an immense network of artificial *canals* built by an intelligent civilization to irrigate a dying world. In support of this argument is the existence of the seasonally varying polar ice caps, visible in the Hubble Space Telescope image shown in Fig. 20.21(a). It is not difficult to imagine that using smaller telescopes, which were peering at the red planet through Earth's obscuring atmosphere, would have led to the conclusion that canali were present. In an effort to verify these features, Percival Lowell (1855–1916) built an observatory near Flagstaff, Arizona, to carry out a series of careful observations of this nearby world. Other astronomers were somewhat more skeptical of the existence of intelligent life on Mars, and even of the canali. However, the general public seized on the possibility that Martians do (or at least did) live there, leading to a wealth of science fiction literature and films.

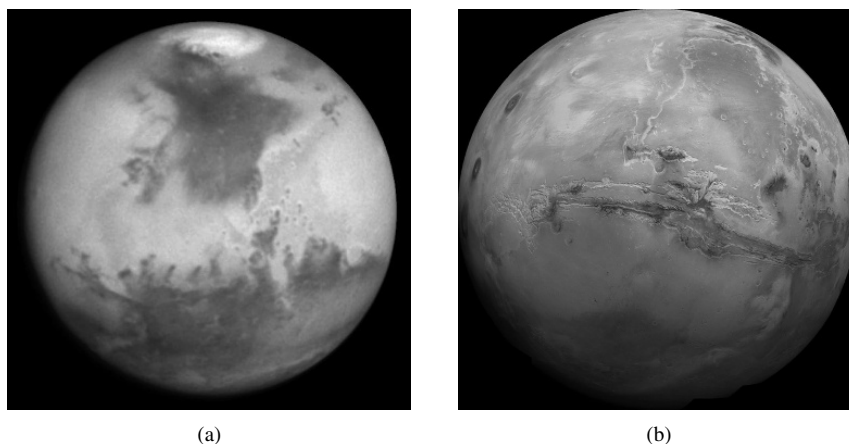


FIGURE 20.21 (a) An image of Mars obtained using WF/PC 2 onboard the Hubble Space Telescope. The north polar ice cap is clearly visible. (Courtesy of Philip James, University of Toledo; Steven Lee, University of Colorado; and NASA.) (b) A mosaic of 102 Viking Orbiter images obtained in 1976. This perspective places the viewer 2500 km above the surface of the red planet. Valles Marineris (a 3000-km-long canyon system) can be seen near the equator. On the left side of the image, three giant shield volcanoes are evident as dark, circular regions. Each volcano is approximately 25 km tall. (Courtesy of U.S. Geological Survey and NASA/JPL.)

Exploration of the Red Planet

There have been many attempts to study Mars by robotic missions. Early efforts included the **Mariner** flyby missions in the 1960s. In 1975 the **Viking** missions included two orbiters and partnered landers that contained cameras and internal laboratories for the study of Mars surface chemistry. Since the landers did not have any ability to move across the surface, their studies were restricted to the locations where they set down on the planet. The **Mars Global Surveyor**, with its very high-resolution camera entered Mars orbit in 1997 and continues to operate successfully at the time of this writing, as does the **Mars Odyssey**, which arrived in 2001, and the **Mars Express Orbiter**, an ESA mission that reached the red planet in 2003. Another spacecraft, the **Mars Reconnaissance Orbiter**, also began its work around the planet in 2006.

The **Sojourner Rover** of the **Mars Pathfinder** mission (1997) was the first truly mobile lander, able to move short distances across the surface in the vicinity of its lander, the **Carl Sagan Memorial Station**.¹⁶ Then in January 2004, two golf-cart-sized rovers successfully landed on the surface and began extensive exploration of the regions around their landing areas. The **Mars Exploration Rovers, Spirit and Opportunity**, were originally expected to operate for several months, but they continued to move across the surface of the planet as late as May 2006. The Mars Orbiter has been able to image both rovers from its vantage point in orbit around the planet. Other missions are also planned, including additional orbiters and landers and possible human-crewed missions to Mars.

¹⁶The stationary lander base was renamed the Carl Sagan Memorial Station after landing in honor of Carl Sagan (1934–1996), Solar System researcher, Pulitzer-prize-winning author, and popularizer of astronomy.

Evidence of Water on Mars

Despite the many studies of Mars from Earth, from Mars orbit, and from its surface, no sign of life has been found on the planet. At first inspection, the images returned by Spirit and Opportunity (see Fig. 20.22), along with images obtained from the Viking landers, give the impression of a dry, dusty world. However, on closer inspection of data returned by Spirit and Opportunity, along with information from the orbiters, research has revealed a fascinating world that, although dry today, once clearly had water flowing across its surface. Apparent in images of the surface from the Mars Orbiter (see Fig. 20.23) are channels that are characteristic of water erosion found on Earth. There is also evidence that huge flash floods may have occurred on the surface of the planet. It appears that lakes of water may have been present on Mars in the distant past as well (Fig. 20.24).

With present-day surface temperatures varying between -140°C (-220°F) and 20°C (70°F), combined with the very low atmospheric pressure found near the surface (typically 0.006 atm), it appears that the liquid water that was present on Mars is now either trapped in a layer of permafrost or frozen in its polar ice caps [Fig. 20.21(a)]. In fact, it is the low atmospheric pressure that makes the existence of persistent water in liquid form impossible on the surface today.

ALH84001, A Martian Meteorite

Ironically, even though the intense investigations of Mars by robotic spacecraft and landers have thus far failed to identify any evidence that life exists on Mars today or existed there in the past, a meteorite discovered in Allan Hills, Antarctica, in 1984 led to speculation that

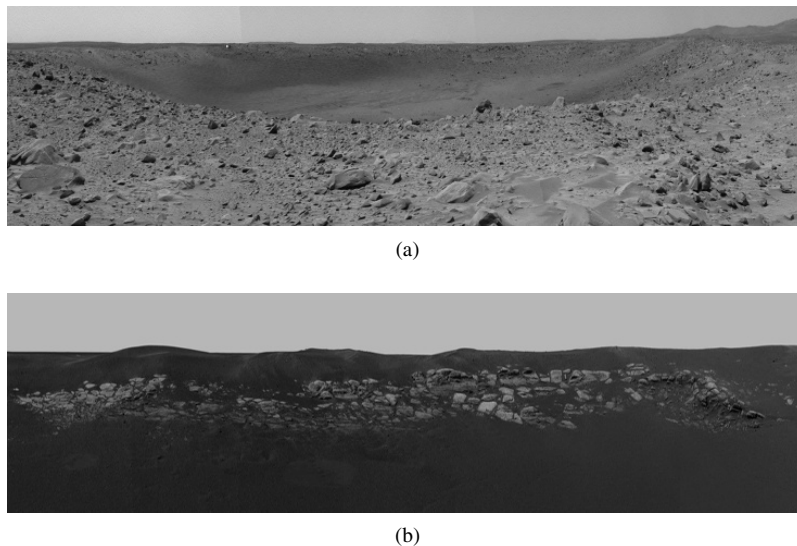


FIGURE 20.22 (a) A panoramic view of the Bonneville crater obtained by Mars Exploration Rover Spirit. (Courtesy of NASA/JPL.) (b) A panoramic view of interesting rock features at Meridiani Planum, near the landing site of Mars Exploration Rover Opportunity. (Courtesy of NASA/JPL.)

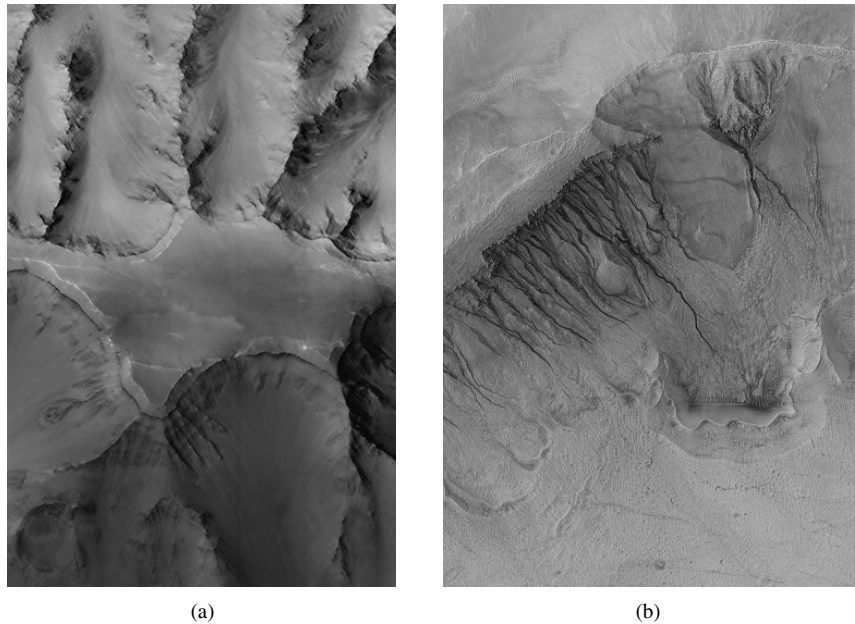


FIGURE 20.23 (a) A portion of Valles Marineris [recall Fig. 20.21(b)] showing evidence of water-caused erosion. (Courtesy of NASA/JPL/Malin Space Science Systems.) (b) Erosion channels seen in an impact crater in Newton Basin in Sirenum Terra, located in the southern hemisphere of Mars. (Courtesy of NASA/JPL/Malin Space Science Systems.)

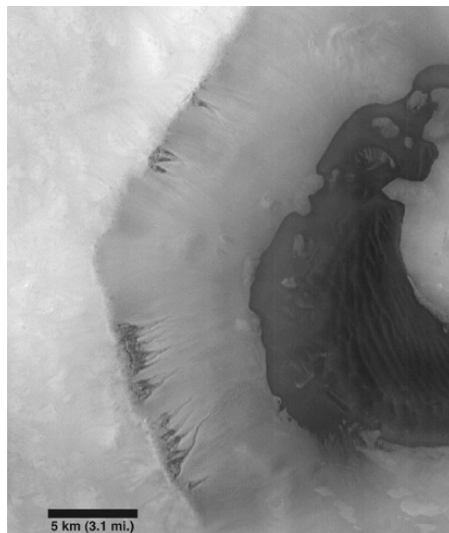


FIGURE 20.24 An impact crater in the southern hemisphere of Mars. The dark material at the bottom is believed to be sediment deposits from an ancient Martian lake. Seepage into the crater is also evident near the rim of the crater. Dunes are visible in the dark regions as well. (Courtesy of NASA/JPL/Malin Space Science Systems.)

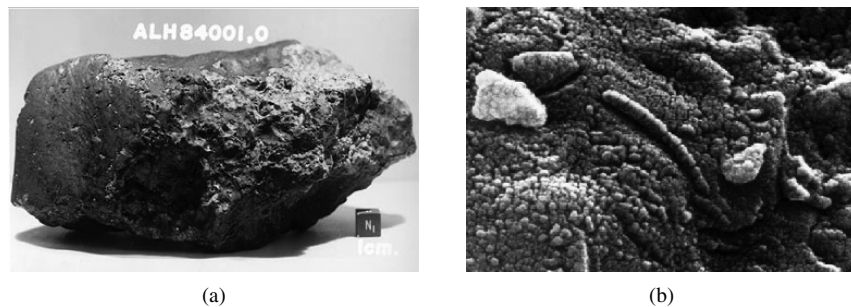


FIGURE 20.25 (a) ALH84001 is a Martian meteorite found in Allan Hills, Antarctica, in 1984. (Courtesy of NASA.) (b) An electron microscopy view of a portion of ALH84001 showing tube-like structures less than 1/100 the size of a human hair. Some scientists have argued that these structures represent nanofossils of ancient microbial life on Mars. (Courtesy of NASA.)

evidence existed in that rock from space [Fig. 20.25(a)]. **ALH84001** is the oldest meteorite ever found that originated from the surface of Mars. After forming on Mars 4.5 billion years ago, it was ejected from the surface of the planet 16 million years ago by an energetic collision. After traveling through the inner Solar System, it struck Earth 13,000 years ago and became trapped in the Antarctic ice sheet.¹⁷ Confirmation that the meteorite did indeed originate on Mars comes from comparing its chemical composition with the results of compositional studies conducted by robotic landers.

It was the examination of small amounts of carbonate grains in the meteorite that has led some researchers to suggest that ancient, fossilized Martian microbes may be contained in the rock [Fig. 20.25(b)]. The grains themselves measure less than $200\ \mu\text{m}$ in size, and what appear to be fossilized microbes are smaller than 1/100 the size of a human hair. In support of the hypothesis that the “nanofossils” are due to ancient microbial life is the presence of organic PAHs in the carbonate, along with oxide and sulfide biominerals. The carbonate grains also appear to have formed in fractures in the rock, possibly in the presence of liquid water.

Most researchers now believe that, although ALH84001 is a fascinating rock that clearly originated on Mars, the evidence is weak that it contains samples of primitive fossilized life. Rather, it could be that the features formed by some inorganic mechanism(s) or that the rock became contaminated as a result of the 13,000 yr it spent on Earth before it was discovered.

The Polar Caps

Although water ice is certainly present today in the polar caps, the caps are composed primarily of dry ice (frozen carbon dioxide). Mars’s axis tilt of 25° and its orbital period of 1.88 yr means that the planet’s seasonal variations are similar to Earth’s but are roughly twice as long. Consequently, Mars experiences winter and summer seasons corresponding to observed variations in the sizes of the ice caps. It is the dry ice that sublimates during the

¹⁷The ejection and landing ages were determined through cosmic ray exposure that the meteoroid received prior to hitting Earth.

Martian summer and freezes back out again during the winter. The small residual cap that remains during the summer is composed of water ice.

Chaotic Fluctuations in Mars's Rotation Axis

From numerical simulations designed to investigate the long-term stability of planetary motions, it appears that the orientation of Mars's spin axis fluctuates wildly (chaotically) between about 0° and 60° over time scales as short as a few million years; the variations are due to gravitational interactions with the Sun and the other planets. If Mars has experienced such large fluctuations in its axis tilt in the past, this would imply that at various times the polar ice caps could completely melt (high tilt angle), whereas at other times the planet's atmosphere might actually freeze out (low tilt angle). The time-variability of the tilt of Mars's spin axis also implies that its current tilt, which is similar to Earth's, is only coincidental.

Interestingly, these simulations of the fluctuations imply that the chaotic behavior does not develop if the effects of general relativity are neglected. It seems that the effects of spacetime curvature discussed in Chapter 17 play an important role in the long-term behavior of planetary orbits and their rotations, even at the distance of Mars's orbit.

Even though it is closer to the Sun, Earth has not experienced the same dramatic oscillations in its axis tilt that Mars seems to have gone through. Apparently Earth's rotation axis is stabilized by our planet's strong tidal interaction with its relatively large moon. Consequently, our planet's climatic variations have been much less pronounced than those on Mars. Amazingly, this seems to imply that the presence of Earth's moon (apparently the result of an accidental collision) is in part responsible for the stable environment that led to the evolution of life on the third planet from the Sun.

The Thin Atmosphere of Mars

Mars's very thin atmosphere is composed of 95% carbon dioxide and 2.7% molecular nitrogen, by number—percentages very similar to those in the atmosphere of Venus. Unlike the case of Venus, however, the greenhouse effect has very little influence on the current equilibrium temperature of Mars; there simply are not enough molecules present to absorb a significant amount of infrared radiation (the atmospheric pressure at the surface of Venus is 90 atm, 13,000 times greater than the atmospheric pressure at the surface of Mars). In the past, the atmosphere of Mars may have been much more dense, causing the greenhouse effect to be more efficient than it is today. The water that is currently trapped in the ice caps and permafrost would then have been flowing freely, maybe even resulting in rainfall. The water that was present in the atmosphere and on the surface would have absorbed much of the atmospheric carbon dioxide, subsequently locking the CO_2 in carbonate rocks. As a result, the greenhouse effect diminished, the global temperature dropped, and the water froze, leaving the dry world we find today.

Shortly after the two Viking landers arrived at Mars in 1975, they began to measure an appreciable drop in atmospheric pressure. This was because winter was coming to the Southern Hemisphere, and carbon dioxide was freezing out of the atmosphere. When spring returned to the south, the atmospheric pressure went back up again. The same behavior was repeated when winter arrived in the Northern Hemisphere.

Dust Storms

Even though the atmospheric density is quite low near the surface, it is sufficient to produce huge dust storms that sometimes cover the entire surface of Mars. The seasonal storms are driven by high winds and are responsible for the variations in surface hues that can be seen from Earth.¹⁸

It was during the Viking missions in 1976 that two such major dust storms occurred. Since that time, much of the dust has settled out of the planet's atmosphere, resulting in noticeable changes in its climate. (The absorption of light by dust is the primary source of atmospheric heating.) In fact, the Hubble Space Telescope recorded a decline in the average global temperature of the planet. With the decrease in average temperature, ice-crystal clouds have become more prominent in the planet's lower atmosphere than they were at the time of the Viking missions.

The Abundance of Iron

The dust on the surface (recall Fig. 20.22) appears reddish in color and contains a relatively high abundance of iron, which oxidizes (rusts) when exposed to the atmosphere. Apparently Mars did not undergo the same degree of gravitational separation that Earth did, possibly because the smaller, more distant planet cooled more rapidly following its formation. However, averaging over the volume of the entire planet reveals that iron is actually underabundant on Mars relative to the other terrestrial planets, as evidenced by its lower average density of 3933 kg m^{-3} . The reason for this is not yet understood.

The lack of significant gravitational separation is also consistent with the absence of an appreciable global magnetic field. If an iron core is present, presumably it is quite small and probably not molten.

Evidence of Past Geological Activity

Even if Mars may not be geologically active today, it certainly has been in the past. Figure 20.21(b) shows Valles Marineris, a 3000-km-long network of canyons near the planet's equator. It appears that Valles Marineris, which is up to 600 km wide in some places and can reach a depth of 8 km, was formed from *faulting* (or fracturing of the crust) in order to relieve stresses that built up in the interior.

Olympus Mons, shown in Fig. 20.26, is a shield volcano that covers an area roughly the size of Utah. The volcano rises 24 km above the surrounding surface and has a huge caldera (a volcanic crater). Geologists believe that Olympus Mons owes its enormous size to a process known as **hot-spot volcanism**, where a weak spot in the crust has allowed molten material to rise to the surface. It is hot-spot volcanism on Earth that is responsible for the creation of the Hawaiian islands.¹⁹ However, in the case of the Hawaiian island chain, motion of the tectonic plate on which the chain rides carries each newly formed volcano away from the hot spot, allowing another one to be created. Today the chain of

¹⁸These seasonal variations were once thought by some astronomers to be evidence of vegetation growing cycles.

¹⁹The tallest mountain on Earth, measured from its base to its summit, is the Hawaiian Island of Mauna Loa, with a vertical rise above the sea floor of 9.1 km.

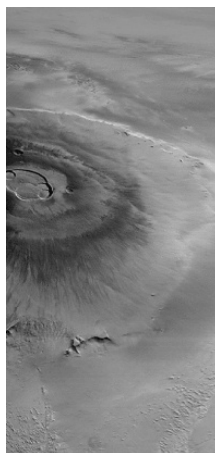


FIGURE 20.26 Olympus Mons is a shield volcano rising 24 km above the surrounding surface. Measured at its base, the diameter of the volcano is more than 500 km. The cliff that rings the volcano in this perspective image is 6 km high. (Courtesy of NASA/JPL.)

mountains that contains the Hawaiian islands actually stretches nearly all the way to Japan, although over time the oldest mountains have undergone significant erosion.²⁰

The situation was somewhat different for Olympus Mons. Since Mars has apparently not developed a system of moving tectonic plates, the volcano was not carried off the hot spot where it formed. As a result, it has grown larger and larger as more molten material has made its way to the surface.²¹

Two Tiny Moons

The two moons of Mars, Phobos and Deimos (Fig. 20.27), were discussed briefly in Section 19.2. Although they were discovered by Asaph Hall (1829–1907) in 1877, Kepler had postulated their existence centuries earlier. His “prediction” was based solely on numerology. Knowing that there were no moons in orbit about Venus, that Earth had one satellite, and that Galileo had recently discovered four moons orbiting Jupiter, Kepler decided it seemed reasonable that Mars ought to have two!

In 1726, 150 years before Hall’s actual discovery, Jonathan Swift (1667–1745) wrote in his book *Gulliver’s Travels* that astronomers had discovered two satellites orbiting the red planet. His fictitious scientists found that the orbital periods of these moons were 10 hours and $21\frac{1}{2}$ hours, “so that the squares of their periodical times are very near in the same proportion with the cubes of their distance from the centre of Mars, which evidently shows them to be governed by the same law of gravitation that influences the other heavenly bodies.” Apparently Swift, who was not a scientist, was aware of scientific discoveries

²⁰The Yellowstone region, with its geysers, hot springs, and mud volcanoes, is another example of hot-spot volcanism on Earth.

²¹The large volcanos discovered on Venus may have been formed in much the same way as Olympus Mons.

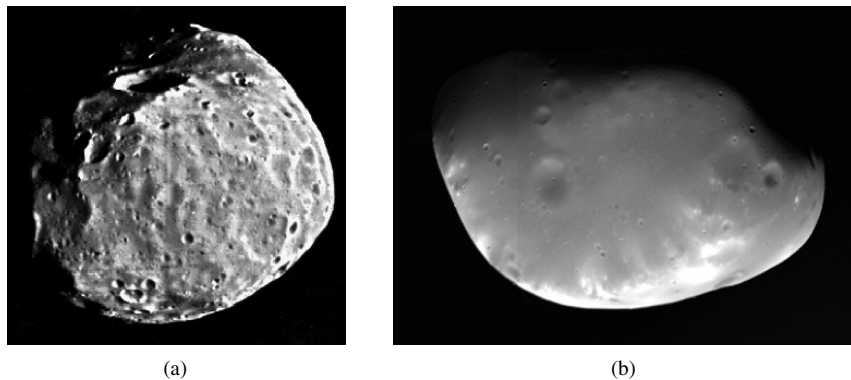


FIGURE 20.27 The two moons of Mars, (a) Phobos and (b) Deimos, are very similar to asteroids and were probably captured by the planet. (Courtesy of NASA/JPL.)

such as Kepler's third law. The actual orbital periods of Phobos and Deimos are $7^{\text{h}}39^{\text{m}}$ and $30^{\text{h}}17^{\text{m}}$, respectively, remarkably close to the values determined by Swift's astronomers.

Both Phobos and Deimos are small, heavily cratered, elongated rocks. Phobos's longest dimension is a mere 28 km, and Deimos's is even smaller (16 km). It appears likely that the moons are captured asteroids.

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Technical

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PROBLEMS

- 20.1** Assume that radar signals of 10 GHz are used to measure the rotation rates of Mercury and Venus. Using the Doppler effect, determine the relative shifts in frequency for signals returning from the approaching and receding limbs of each planet.
- 20.2** What is the ratio of the Sun’s tidal force per unit mass on Mercury at perihelion to the Sun’s tidal force per unit mass on Earth? How has this difference in tidal effects contributed to differences in the orbital and/or rotational characteristics of the two planets?
- 20.3** For Mercury, a slowly rotating planet with no appreciable atmosphere, Eq. (19.5) for a planet’s surface temperature must be modified. In particular, the assumption that the temperature is approximately constant over the entire surface of the planet is no longer valid.
- (a) Assuming (incorrectly) that Mercury is in synchronous rotation about the Sun, show that the temperature at a latitude θ north or south of the *subsolar point* (the point on the equator closest to the Sun) is given by

$$T = (\cos \theta)^{1/4} (1 - a)^{1/4} T_{\odot} \sqrt{\frac{R_{\odot}}{D}}.$$

Since the planet is actually in a 3-to-2 resonance, this expression is only an approximate description for the temperature at Mercury’s surface.

- (b) Make a graph of T vs. θ . Mercury’s albedo is 0.06.
- (c) What is the approximate temperature of the planet at the subsolar point?
- (d) At what latitude does the temperature drop to 273 K? This is the freezing point of water at the surface of Earth.
- (e) Would you expect to find ice on Mercury at a temperature of 273 K? Why or why not?
- 20.4** (a) Estimate the angular resolution of the 70-m radio dish of the NASA Goldstone tracking station mentioned in footnote 1 on page 739. Assume that it is operating at a wavelength of 3.5 cm.
- (b) What is the angular size of Mercury at inferior conjunction? Assume (incorrectly) for this problem that the planet’s orbit is circular.
- (c) If the power in the radar signal was approximately uniformly distributed across the cone-shaped beam, how much power actually arrived at the surface of Mercury?

- (d) Suppose that all of the radar energy striking the surface of Mercury were reflected isotropically back into a hemisphere. What would be the signal flux received at the VLA?
- 20.5** (a) From the data presented in the text, estimate the kinetic energy of the impact that may have been responsible for stripping off the outer layers of Mercury early in the history of the Solar System.
- (b) If, prior to the collision, Mercury had twice as much mass as it does today, how much energy would have been required to lift that additional mass off the present planet? Assume that the extra mass had the density of Earth's present-day Moon and that the material was uniformly distributed in a spherical shell around the present-day Mercury. Don't forget to include the energy required to eject the mass of the impactor as well.
- (c) Solely on the basis of energy considerations comment on the feasibility of this scenario for the origin of Mercury as we observe it today.
- 20.6** Assuming that the atmosphere of Venus is composed of pure carbon dioxide, estimate the number density of molecules at the planet's surface. How many times larger is this value than the number density of nitrogen molecules at the surface of Earth, as quoted in Example 19.3.3?
- 20.7** (a) Modeling the greenhouse effect using one atmospheric layer, as was done in Problem 19.13, is equivalent to assuming that the optical depth is about one. If the optical depth is τ , and if we can neglect circulation in the atmosphere, show that the surface temperature should be approximately

$$T_{\text{surf}} = (1 + \tau)^{1/4} T_{\text{bb}},$$

where T_{bb} is the blackbody temperature of an *airless* planet.

- (b) The optical depth of Venus's atmosphere is approximately $\tau = 70$. Make an estimate of its surface temperature using this crude greenhouse model. Take the average albedo to be 0.77.
- 20.8** Based on the observed rate at which North America and Eurasia are separating from each other, when were the two continents joined together as Laurasia? Assume that the Atlantic Ocean is roughly 4800 km (3000 miles) wide.
- 20.9** Using the equation of hydrostatic equilibrium (Eq. 10.6), estimate the pressure at the center of Earth. Detailed computer simulations suggest that the central pressure is 3.7×10^6 atm.
- 20.10** (a) From the data given in Section 19.2, estimate the *rate* at which rotational energy is being dissipated by tidal friction for the case of Earth. *Hint:* This terrestrial problem is similar to the loss of rotational kinetic energy in pulsars; see Eq. (16.30).
- (b) What fraction of the total energy being lost from Earth's interior can be accounted for by tidal dissipation of its rotational kinetic energy?
- 20.11** Referring to Eq. (11.2) for the Lorentz force and Fig. 20.14, explain why most charged particles bounce back and forth between the North and South Poles of Earth, rather than striking the surface. Use a diagram if necessary. *Hint:* The converging magnetic field lines form magnetic *mirrors* near the North and South Poles. (Magnetic "bottles," which are based on the same principle, are used to confine high-temperature plasmas in laboratories.)
- 20.12** The moment of inertia of a planet is used to evaluate its interior structure. In this problem you will construct a simple "two-zone" model of the interior of Earth, assuming spherical symmetry. Take the average densities of the core and mantle to be $10,900 \text{ kg m}^{-3}$ and 4500 kg m^{-3} , respectively (neglect the thin surface crust).

- (a) Using the average density of the entire Earth, determine the radius of the core. Express your answer in units of Earth's radius.
- (b) Calculate the *moment-of-inertia ratio* (I/MR^2) for the “two-zone” Earth (the actual value is 0.3315). The moment of inertia for a spherically symmetric mass shell of constant density ρ , having inner and outer radii R_1 and R_2 , respectively, is given by

$$I \equiv \int_{\text{vol}} a^2 dm = \frac{8\pi\rho}{15} (R_2^5 - R_1^5).$$

a is the distance of the mass element dm from the axis of rotation.

- (c) Compare your answer in part (b) with the value expected for a solid sphere of constant density. Why are the two values different? Explain.
- 20.13** The moment-of-inertia ratio of the Moon is 0.390 (see Problem 20.12).
- (a) What does this say about the interior of the Moon?
- (b) Is this consistent with the lack of any detectable magnetic field? Why or why not?
- 20.14** (a) The *Apollo 11* astronauts, after landing on the Moon on July 20, 1969, returned rocks from the Sea of Tranquility, one of the maria on the near side. Upon their return, the analysis of one rock (basalt 10072) yielded the relative abundances at various locations in the sample; see Table 20.2. Graph the abundance data as $^{143}_{60}\text{Nd}/^{144}_{60}\text{Nd}$ vs. $^{147}_{62}\text{Sm}/^{144}_{60}\text{Nd}$. (Note that the uncertainties listed correspond to the last two significant figures.)
- (b) Determine the slope of the best-fit straight line drawn through the data and estimate the age of the lunar sample. Compare your answer with the age of the lunar highland sample, determined in Example 20.4.1 from the data in Fig. 20.19.
- 20.15** Estimate the initial rotation period of Earth if the Moon were torn from it, as suggested by the fission model.
- 20.16** Estimate the Roche limit for the Earth–Moon system. Express your answer in units of the radius of Earth. Is the Moon in any danger of becoming tidally disrupted?
- 20.17** Mars is at its closest approach to the Sun during the summer months in its southern hemisphere.
- (a) Using Eq. (19.5), estimate the ratio of the average temperatures on Mars when it is at perihelion and aphelion.

TABLE 20.2 Results from the Analysis of Basalt 10072, Returned from the Sea of Tranquility by the *Apollo 11* Astronauts in 1969. (Data from D. A. Papanastassiou, D. J. DePaolo, and G. J. Wasserburg, “Rb–Sr and Sm–Nd Chronology and Genealogy of Mare Basalts from the Sea of Tranquility,” *Proceedings of the Eighth Lunar Science Conference*, Pergamon Press, New York, 1977.)

$^{147}_{62}\text{Sm}/^{144}_{60}\text{Nd}$	$^{143}_{60}\text{Nd}/^{144}_{60}\text{Nd}$
0.1847	0.511721 $\pm$ 18
0.1963	0.511998 $\pm$ 16
0.1980	0.512035 $\pm$ 21
0.2061	0.512238 $\pm$ 17
0.2715	0.513788 $\pm$ 15
0.2879	0.514154 $\pm$ 17

- (b) Considering the tilt of the planet's rotation axis, describe the seasonal behavior of the two polar ice caps.
- 20.18** Assuming that the two moons are in circular orbits, determine the orbital radii of Phobos and Deimos. Express your answers in units of the radius of Mars.
- 20.19** Suppose you lived on Mars and watched its moons. If Phobos and Deimos were next to each other one night, what would you see the next night (one Martian day later)? Describe the apparent motions of the two moons. (Both Phobos and Deimos orbit prograde, approximately above the planet's equator.)

CHAPTER

21

The Realms of the Giant Planets

- 21.1 *The Giant Worlds*
- 21.2 *The Moons of the Giants*
- 21.3 *Planetary Ring Systems*

21.1 ■ THE GIANT WORLDS

Excluding the Sun, by far the largest member of the Solar System is **Jupiter**, 317.83 times more massive than Earth. Jupiter and the other three giants, **Saturn**, **Uranus**, and **Neptune**, together contain 99.5% of the entire mass of the planetary system (see Fig. 21.1). Consequently, if we hope to understand the development and evolution of our Solar System, it is vital that we understand these distant worlds.

The Discovery of the Galilean Moons

Naked-eye observations of Jupiter and Saturn began when human beings first started gazing up at the heavens. But it was in 1610 that Galileo became the first person to look at these planets through a telescope. In so doing, he detected the four large moons of Jupiter, now collectively known as the **Galilean moons**.¹ Galileo also saw Saturn's rings, but because of his telescope's low resolution, he thought that the rings were two large satellites situated on either side of the planet.

The Discoveries of Uranus and Neptune

It wasn't until 1781 that William Herschel (1738–1822), a German-born musician living in England, made the chance discovery of Uranus. By considering gravitational perturbations affecting the orbit of Uranus, John Couch Adams (1819–1892), a graduate student at Cambridge University, proposed in October 1845 that another planet must exist even farther from the Sun. Using Bode's rule to guess at the distance of this unknown planet from the Sun, Adams predicted its position in the heavens. Unfortunately, when he submitted his work to Sir George Airy, the Astronomer Royal of England, Airy did not believe the conclusions. In June 1846, Urbain Leverrier (1811–1877), a very well-respected French scientist, independently made the same prediction, agreeing with Adams's position to within 1°. Learning of the agreement between the two predictions, Airy began to search for the object. However,

¹The four Galilean moons were also discovered independently by Simon Marius (1570–1624) in 1610.

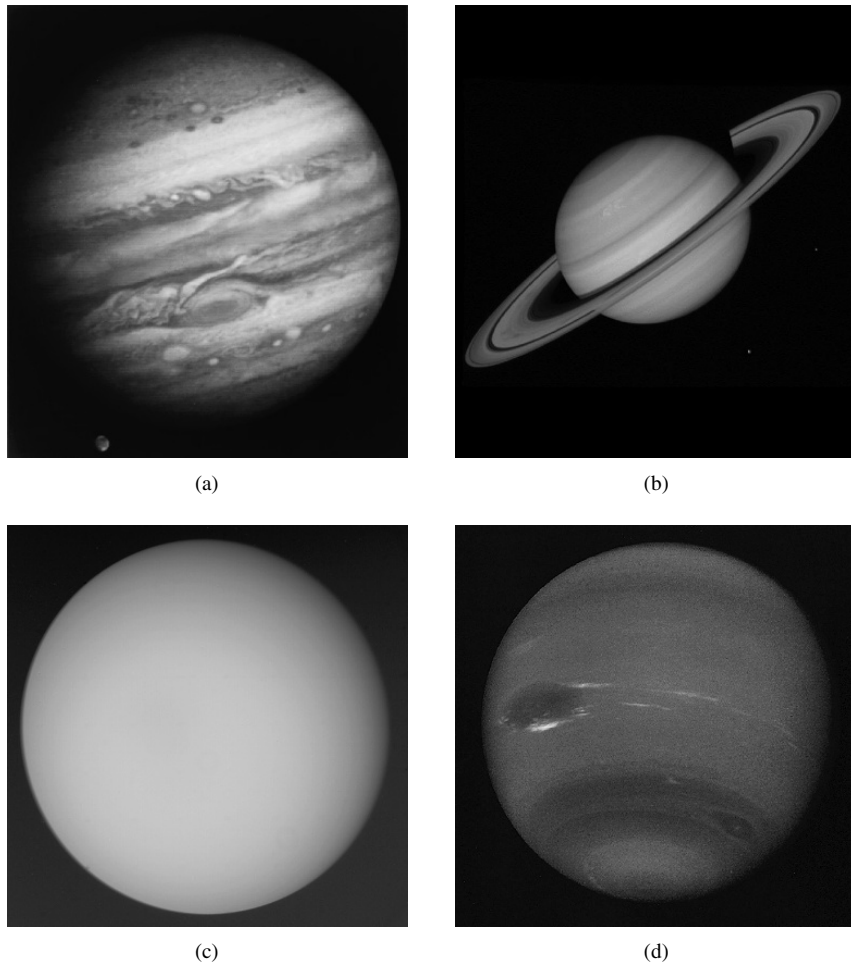


FIGURE 21.1 The four giant planets. (a) Jupiter and its largest moon, Ganymede. (b) Saturn, with two of its moons, Rhea and Dione, seen near the bottom and right-hand side of the image, respectively. (c) Uranus. (d) Neptune. The images were taken by the Voyager 1 and 2 spacecraft. Notice the oblateness caused by rapid rotation. The image sizes do not correspond to the actual relative sizes of the planets. (Courtesy of NASA/JPL.)

Johann Gottfried Galle (1812–1910) of the Berlin Observatory found Neptune on September 23, 1846, the night after receiving a letter from Leverrier suggesting that he too should look for this new planet. In a very real sense, Neptune was discovered in the mathematical calculations of Adams and Leverrier; Galle merely confirmed their work.

Missions to the Giant Planets

Since the first observations of these worlds, the efforts of Earth-based astronomers have provided important information about the giant planets and their many satellites. However, many of the data now available have come from spacecraft missions. The first such missions

were the **Pioneer 10** and **Pioneer 11** flybys of Jupiter (1973, 1974) and the Pioneer 11 flyby of Saturn (1979). Later, **Voyager 1** and **Voyager 2** embarked on their spectacularly successful “Grand Tour” missions. Both Voyagers, launched from Earth in 1977, visited Jupiter (1979) and Saturn (1980, 1981), and Voyager 2 continued on to Uranus (1986) and Neptune (1989). In each case the encounters with the planets were brief flybys. Today the Pioneer² and Voyager spacecraft are on their way out of the Solar System. The Voyager spacecraft (renamed the *Voyager Interstellar Missions*) continue to send back information over immense distances with ever-weakening signals, providing data about the outer reaches of the Solar System, including the interaction between the solar wind and the winds from other stars. In early 2006, Voyager 1 was 8.7 billion miles (14 billion kilometers) from Earth, traveling at a speed of 3.6 AU per year, and Voyager 2 was 6.5 billion miles (10.4 billion kilometers) away, traveling at a speed of 3.3 AU per year. It is believed that Voyager 1 passed the solar wind’s termination shock in December 2004, as evidenced by an increase in the strength of the magnetic field in the vicinity of the spacecraft by a factor of 2.5 (recall Example 11.2.1).

The Hubble Space Telescope has also been used to observe the outer planets from Earth orbit. HST has documented significant changes in the planets since the flyby missions of the 1970s and 1980s.

An extended and detailed investigation of the Jovian system began in 1995 when the **Galileo** spacecraft (launched in 1989) entered into orbit around Jupiter. In addition to observing the planet carefully, Galileo completed numerous flybys of the Galilean moons during the eight years it spent in the Jovian system. As a part of the mission, a probe descended into the planet’s atmosphere by parachute, sampling the atmosphere’s composition and physical conditions.

The **Cassini–Huygens** mission, launched in 1997, entered the Saturnian system July 1, 2004. This dual mission is composed of the Cassini orbiter, which was built by NASA with the high-gain antenna system provided by the Italian Space Agency (ASI), and the Huygens probe, which was built by the European Space Agency (ESA). At the time of this writing in 2006, Cassini is exploring the Saturnian system at length during its four-year mission, including the planet, its moons, and its rings. Huygens descended into the thick atmosphere of Titan, the largest of Saturn’s moons on January 14, 2005. Like the Galileo probe, Huygens used a parachute during part of its descent while it made measurements of composition, wind speed, atmospheric structure, and surface features. At an altitude of 40 km, the parachute was released and the probe fell to the surface. The descent took two hours and 27 minutes, and the probe remained operational on the surface for an additional one hour and 10 minutes, while further observations were made.

Composition and Structure

Referring to Table 19.1, note that as a class, the giant planets differ markedly from the terrestrials. However, the group can be further subdivided. The gas giants of Jupiter ($317.83 M_{\oplus}$) and Saturn ($95.159 M_{\oplus}$) have average compositions that are quite similar to the Sun’s,

²The last signal received from Pioneer 10 occurred on January 23, 2003, almost 31 years after the spacecraft’s launch. Pioneer 10 is now more than 8 billion miles from Earth, headed in the general direction of Aldebaran, in the constellation of Taurus, and will be in that star’s vicinity in about 2 million years. Pioneer 11 was last heard from in 1995 and is headed in the direction of Aquila.

while the much smaller and more distant ice giants, Uranus ($14.536 M_{\oplus}$) and Neptune ($17.147 M_{\oplus}$), have higher proportions of heavier elements. Because each of the giant planets is capable of retaining all of the lighter elements in its atmosphere, this composition difference suggests important differences in their formation.

This conclusion is supported by direct observations of the composition of each giant planet near the cloud tops. Table 21.1 gives the relative number densities of constituents in the giant planet atmospheres; the Sun's photospheric composition is given for comparison. (Note that it is the percentage of atoms or molecules by *number* that is being quoted, not the *mass fraction* that was discussed in Chapter 10.) Jupiter's hydrogen content is somewhat greater than the Sun's, while its helium content is slightly less than solar. Saturn's upper atmosphere is noticeably deficient in helium (96% H_2 , 3% He), while the other percentages are similar to those of Jupiter. Observations also indicate that whereas their hydrogen and helium contents are intermediate between the Sun and Jupiter, the atmospheres of both Uranus and Neptune are overabundant in methane relative to solar by a factor of 10 or more. Although these studies hint that differences may exist in the interiors of these planets, other observational data and theoretical investigations give us even more information about what is going on inside.

Figure 21.2 shows each planet's radius as a function of its mass. Also plotted are a series of theoretical curves for various mixtures: "H" for pure hydrogen; "H-He" for a hydrogen-helium mixture appropriate for Jupiter and Saturn; "Ice" for a composition of H_2O (water), CH_4 (methane), and NH_3 (ammonia) ice; and "Rock" for a composition of magnesium, silicon, and iron. The dashed lines correspond to models that follow adiabatic temperature gradients. In particular, the gas models (H, H-He) incorporate the polytropic relationship, $P \propto \rho^2$, appropriate for Coulomb-force pair interactions (recall the discussion of polytropes beginning on page 334). $P \propto \rho^2$ is a reasonable approximation when electron-ion pair interactions are important, because $F \propto q^2$ and the number of charges is proportional to the density of the gas. The solid lines in Fig. 21.2 represent zero-temperature models, corresponding to complete degeneracy (see the discussion of building completely degenerate white dwarf models in Section 16.3). It seems that hydrogen and helium dominate in Jupiter and Saturn, while ices are likely to play key roles in determining the interior structures of Uranus and Neptune.

Notice in Fig. 21.2 that even though Jupiter is more than three times as massive as Saturn, it is only slightly larger than its more distant neighbor. This is because the increased mass

TABLE 21.1 Composition of the Atmospheres of the Giant Planets. All values are given as a fractional number density of particles. Jupiter data are from the Galileo probe. Solar photospheric data are provided for comparison. (Data from Table 4.5 of de Pater and Lissauer, *Planetary Sciences*, Cambridge University Press, Cambridge, 2001.)

Gas	Sun	Jupiter	Saturn	Uranus	Neptune
H_2	H: 0.835	0.864 ± 0.006	0.963 ± 0.03	0.85 ± 0.05	0.85 ± 0.05
He	He: 0.195	0.157 ± 0.004	0.034 ± 0.03	0.18 ± 0.05	0.18 ± 0.05
H_2O	O: 1.70×10^{-3}	2.6×10^{-3}	$> 1.70 \times 10^{-3}?$	$> 1.70 \times 10^{-3}?$	$> 1.70 \times 10^{-3}?$
CH_4	C: 7.94×10^{-4}	$(2.1 \pm 0.2) \times 10^{-3}$	$(4.5 \pm 2.2) \times 10^{-3}$	0.024 ± 0.01	0.035 ± 0.010
NH_3	N: 2.24×10^{-4}	$(2.60 \pm 0.3) \times 10^{-4}$	$(5 \pm 1) \times 10^{-4}$	$< 2.2 \times 10^{-4}$	$< 2.2 \times 10^{-4}$
H_2S	S: 3.70×10^{-5}	$(2.22 \pm 0.4) \times 10^{-4}?$	$(4 \pm 1) \times 10^{-4}?$	$3.7 \times 10^{-4}?$	1×10^{-3}

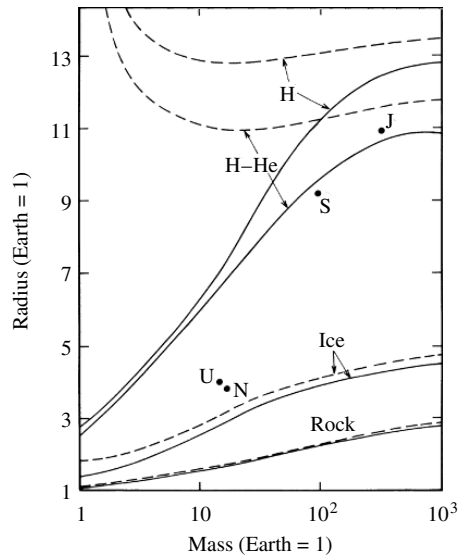


FIGURE 21.2 Composition and mass are principal elements in determining the radius of a planet. Plotted are the radii of Jupiter (J), Saturn (S), Uranus (U), and Neptune (N) as functions of their masses. Also shown are theoretical curves for various mixtures. The solid lines represent zero-temperature models, and the dashed lines are models that follow adiabatic temperature gradients. (Figure adapted from Stevenson, *Annu. Rev. Earth Planet. Sci.*, 10, 257, 1982. Reproduced with permission from the *Annual Review of Earth and Planetary Sciences*, Volume 10, ©1982 by Annual Reviews Inc.)

results in increased interior pressure, which in turn leads to changes in the state of the atoms and molecules. [Recall the equation of hydrostatic equilibrium, Eq. (10.6), developed for stellar interiors; it applies to spherically symmetric planets as well.] For models of objects just over three times more massive than Jupiter and having similar compositions, increasing mass actually results in *decreasing* radius, an effect that begins to appear in the solid H–He curve in Fig. 21.2. This is due to the growing contribution of degenerate electron pressure in these cold, massive bodies. (This counterintuitive behavior of degenerate matter was discussed in some detail in Section 16.4 in the context of the sizes of white dwarf stars.)

The Distribution of Mass Inside the Planets

Other information concerning the distribution of mass in the interior is obtained by observing the motions of moons, rings, and spacecraft. For a spherically symmetric planet, all of the mass acts gravitationally as if it is located at a point in the center, but a rapidly rotating planet produces a more complex gravitational interaction with passing objects. By comparing the actual motion of a spacecraft with what would be expected if the planet were spherically symmetric, it becomes possible to map the mass distribution in the interior in terms of mathematical corrections to a spherical shape. This is just what was done using the Magellan spacecraft around Venus, as discussed in Section 20.2.

One such correction is the **oblateness** of the planet, which describes how flattened it is. Such rotational flattening is readily apparent in Fig. 21.1. For instance, Jupiter’s equatorial radius (R_e) is 71,493 km and its polar radius (R_p) is only 66,855 km at an atmospheric

pressure of 1 bar,³ giving an oblateness of

$$b \equiv \frac{R_e - R_p}{R_e} = 0.064874.$$

The amount of oblateness is a function of the speed of rotation and the rigidity of the interior. The rotation period and oblateness of each of the giant planets are given in Appendix C. Note, however, that since the giant planets are fluid throughout much of their interiors, it is not possible to define a single, unique rotation period; their upper atmospheres tend to rotate *differentially*, just as the Sun does (page 364), while their interiors may rotate at different rates than their surfaces.

The oblateness is related to the first-order correction term in the gravitational potential (the potential energy per unit mass), defined to be

$$\Phi \equiv \frac{U}{m}.$$

For a spherically symmetric mass distribution, $\Phi = -GM/r$, where r is the distance from the center of the planet. However, for a planet that is not exactly spherically symmetric, the gravitational potential can be expanded as an infinite series of the form

$$\Phi(\theta) = -\frac{GM}{r} \left[1 - \left(\frac{R_e}{r} \right)^2 J_2 P_2(\cos \theta) - \left(\frac{R_e}{r} \right)^4 J_4 P_4(\cos \theta) - \dots \right], \quad (21.1)$$

where each succeeding correction term represents a progressively higher-order component of the planet's shape and mass distribution, much like higher-order terms in the familiar Taylor series. Notice that as r increases, each successive higher-order term becomes less significant; as $r \rightarrow \infty$, Φ approaches the form of the spherical potential.

The functions $P_2, P_4, \dots$ are known as **Legendre polynomials** and are encountered frequently in many areas of physics. Each polynomial has $\cos \theta$ as its argument, where θ is the angle between the rotation axis and the position vector of a point in space (the origin of the coordinate system is centered in the middle of the planet); see Fig. 21.3. Examples of some low-order, even-powered Legendre polynomials are

$$P_0(\cos \theta) = 1$$

$$P_2(\cos \theta) = \frac{1}{2} (3 \cos^2 \theta - 1)$$

$$P_4(\cos \theta) = \frac{1}{8} (35 \cos^4 \theta - 30 \cos^2 \theta + 3)$$

$$P_6(\cos \theta) = \frac{1}{16} (231 \cos^6 \theta - 315 \cos^4 \theta + 105 \cos^2 \theta - 5).$$

The Legendre polynomials are multiplied by *weighting factors*, known as **gravitational moments** ($J_2, J_4, J_6, \dots$), that describe the importance of each polynomial to the overall

³1 bar = 10^5 N m^{-2}